

ELE 215.3 Network Theory (3-1-2)

Evaluation:

	Theory	Practical	Total
Sessional	30	20	50
Final	50	-	50
Total	80	20	100

Course Objectives:

The purpose of the course is to provide the knowledge of network equations and the behavior of network. Moreover, it provides in-depth knowledge to develop one-port and two port networks with given network functions.

Course Contents:

1. Review of Network Analysis (2 hrs)
 - 1.1 Mesh and Nodal analysis
2. Circuit Differential Equations (Formulation and Solutions) (5 hrs)
 - 2.10 The differential operator
 - 2.11 Operational impedance
 - 2.12 Formulation of circuit differential equations
 - 2.13 Complete response (transient and steady state) of first order differential equations with or without initial conditions
3. Circuit Dynamics (7 hrs)
 - 3.7 First order RL and RC circuits
 - 3.8 Complete response of RL and RC circuit to sinusoidal input
 - 3.9 RLC circuit
 - 3.10 Step response of RLC circuit
 - 3.11 Response of RLC circuit to sinusoidal inputs
 - 3.12 Damping factors and Damping Coefficients.
4. Review of Laplace Transform (5 hrs)
 - 4.7 Definition and properties
 - 4.8 Laplace transform of common forcing functions
 - 4.9 Initial and final value theorem
 - 4.10 Inverse Laplace transform
 - 4.11 Partial fraction expansion
 - 4.12 Step response of RL, RC and RLC circuit

- 4.13 Sinusoidal response of RL, RC and RLC circuits
4.14 Exponential response of RL, RC and RLC circuits

(4 hrs)

5. Transfer Functions

- 5.7 Transfer functions of network system
5.8 Poles and Zeros
5.9 Time domain behavior from pole-zero locations
5.10 S Routh's Hurwitz's stability Criteria

(4 hrs)

6. Fourier Series and Transform

- 6.8 Evaluation of Fourier coefficients for periodic non-sinusoidal waveform
6.9 Fourier Transform
6.10 Application of Fourier transforms for non-periodic waveforms

(7 hrs)

7. Frequency Response of Network

- 7.10 Magnitude and phase responses
7.11 Bode plots and its applications
7.12 Concept of ideal and non-ideal low pass, high pass, band pass, and band reject filters

8. One-port Passive Network

(7 hrs)

- 8.12 Properties of one-port passive network
8.13 Positive Real Function
8.14 Properties of RL, RC and LC network
8.15 Synthesis of RL, RC and LC networks using Foster's and Cauer's method
8.16 Properties of RLC one-port network

9. Two-port Passive Network

(7 hrs)

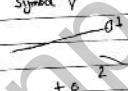
- 9.8 Properties of two-port network
9.9 Reciprocity and symmetry
9.10 Short circuit and Open circuit parameters
9.11 transmission parameters
9.12 Hybrid parameter
9.13 Relation and transformations between sets of parameters
9.14 Equivalent T and π section representation

Laboratory:

1. Transient and steady state responses of first order Passive network
 - 1.1 Measurement of step, impulse and ramp response of RC and RL circuit using oscilloscope
 - 1.2 Measurement of sinusoidal response of RC and RL circuit using oscilloscope

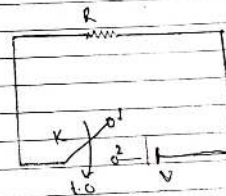
Network Theory

Switch concept [30 marks]

Symbol $\rightarrow$
open $\rightarrow$ 
Switch close if 1 connects with 2.

$t=0^-$ $t=0$ $t=0^+$
Initial condition $\rightarrow$ immediate after switching
final condition.

Resistor (R):



Switch K is moved from position 1 to 2 at time $t=0$

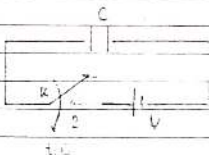
$i(0^-) = 0$
Initial value

After switching

$$i(0^+) = \frac{V}{R}$$

$i(0^-) \neq i(0^+)$ for Resistor

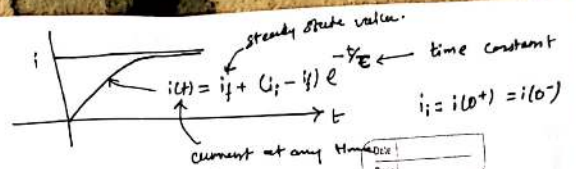
Capacitor (C)



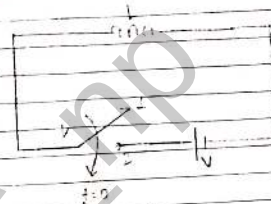
charge = Q
voltage = V

$$Q(0^-) = 0 = Q(0^+)$$

$V_C(0^-) = V_C(0^+)$
initial voltage



Inductor (L)



$$i(0^-) = i(0^+)$$

Differential Equation

$$i(t) = i_{\infty} + (i_0 - i_{\infty})e^{-t/\tau}$$

$$L \frac{di}{dt} + Ri = V \dots (1)$$

Order = 1

degree = 1

$$\frac{di}{dt} + \frac{R}{L}i = \frac{V}{L}$$

Homogenous Equation if $v=0$

$$\frac{di}{dt} + \frac{R_1}{L}i = 0$$

$$\frac{di}{dt} = -\frac{R_1}{L}i$$

$$\frac{di}{i} = -\frac{R_1}{L} dt$$

Integrating both side

$$\int \frac{1}{i} di = \int -\frac{R_1}{L} dt$$

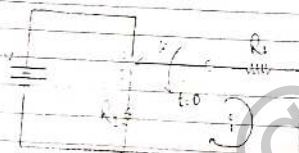
$$\ln i = -\frac{R_1}{L}t + k$$

$$i = K e^{-\frac{R_1}{L}t}$$

Step Function Response for R-L series circuit

The switch is moved from position 1 to 2 at $t=0$

Q find the response in current at $t=0^+$



Solution

KVL eqn at $t=0$

$$R_1i + R_2i + L \frac{di}{dt} = 0$$

$$\text{or } (R_1 + R_2)i + L \frac{di}{dt} = 0$$

$$\text{or } \frac{di}{dt} + \frac{(R_1 + R_2)}{L}i = 0 \quad \text{1st order diff. eqn}$$

$$\frac{di}{i} = -\frac{(R_1 + R_2)}{L} dt$$

$$\text{or } i = K e^{-\frac{(R_1 + R_2)}{L}t} \quad \text{--- (1) General eqn}$$

To find K , at $t=0$

As we know $i(0^-) = \frac{V}{R_1} = i(0^+)$

$$i(0^+) = \frac{V}{R_1}$$

By nature of inductor, it carries the same initial current just after switching, i.e. $i(0^-) = i(0^+) = \frac{V}{R_1}$

Put in eqn (1), $\frac{(R_1 + R_2)}{L} = 0$

$$\frac{V}{R_1} = K e^{-\frac{(R_1 + R_2)}{L}t}$$

$$\text{or } K = \frac{V}{R_1}$$

So from (1) we get

$$i(0^+) = \frac{V}{R_1} e^{-\frac{(R_1+R_2)}{L}t}$$

This is the complete response.

for Inductor

$$V = L \frac{di}{dt}$$

$$V dt = L di$$

$$di = \frac{V}{L} dt$$

$$\therefore i = \frac{1}{L} \int V dt$$

for capacitor

$$V = \frac{Q}{C}$$

$$\text{or, } V = \frac{1}{C} \int \frac{di}{dt} dt$$

$$CV = \int i dt$$

$$i = \frac{C dv}{dt}$$

$$R_1 = 5\Omega$$

$$R_2 = 10\Omega$$

$$L = 0.1H$$

$$V = 20V$$

find the current at $t = 10\text{ sec}$.

Solⁿ

$$\text{We've } i(0^+) = \frac{V}{R_1} e^{-\frac{(R_1+R_2)}{L}t}$$

$$= \frac{20}{5} e^{-\frac{(5+10)}{0.1} \times 10}$$

$$= \frac{20}{5} e^{-2000}$$

$$= 0$$

Solution for non homogeneous equation

$$\text{let eqn is } \frac{di}{dt} + \alpha_0 i = \alpha_1$$

$$\alpha_0 \frac{di}{dt} + \alpha_0 i = V$$

$$\text{or, } \frac{di}{dt} + \frac{\alpha_0}{\alpha_0} i = \frac{V}{\alpha_0}$$

$$\text{or, } \frac{di}{dt} = \frac{V - \alpha_0 i}{\alpha_0}$$

$$\text{or, } \int \frac{di}{V - \alpha_0 i} = \int \frac{1}{\alpha_0} dt$$

$$\text{or, } -\frac{1}{\alpha_0} \ln(V - \alpha_0 i) = \frac{1}{\alpha_0} t$$

$$\text{or, } V - \alpha_0 i = e^{-\frac{\alpha_0}{V} t} \Rightarrow i = \frac{V - e^{-\frac{\alpha_0}{V} t}}{\alpha_0}$$

Solution of first order non-homogeneous equation.

Let the equation is

$$\frac{di}{dt} + Ri = \mathcal{E} \dots \textcircled{1}$$

Multiplying both sides by e^{Rt} known as integrating factor.

$$e^{Rt} \frac{di}{dt} + e^{Rt} Ri = e^{Rt} \mathcal{E} \dots \textcircled{2}$$

$$\text{or, } d(i e^{Rt}) = \mathcal{E} e^{Rt} dt$$

$$\text{let } x = i$$

$$y = e^{Rt}$$

$$\frac{d(i e^{Rt})}{dt} = e^{Rt} \frac{di}{dt} + i R e^{Rt} \dots \textcircled{3}$$

Equating equation ② and ③

$$\frac{d(i e^{Rt})}{dt} = e^{Rt} \mathcal{E}$$

Integrating both sides,

$$i e^{Rt} = \int e^{Rt} \mathcal{E} dt + K$$

$$\text{or, } i(t) = \frac{1}{e^{Rt}} \int e^{Rt} \mathcal{E} dt + \frac{K}{e^{Rt}}$$

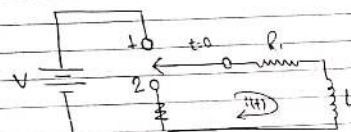
$$\text{or, } i(t) = e^{-Rt} \int \mathcal{E} e^{Rt} dt + K e^{-Rt}$$

First term = $e^{-Rt} \int \mathcal{E} e^{Rt} dt$ (Particular integral) steady state

Second term = $K e^{-Rt}$ (Complementary function) transient state

$$i(t) = PI + CF$$

* Switch K moved from 2 to 1 at time $t=0$ find the current response $i(t)$ at $t=0^+$



Soln

At $t=0$

Using KVL

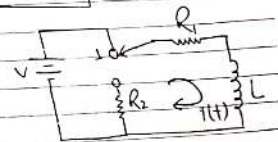
$$-R_1 i - L \frac{di}{dt} + V = 0$$

$$\text{or, } L \frac{di}{dt} + R_1 i = V$$

$$\frac{di}{dt} + \left(\frac{R_1}{L}\right) i = \frac{V}{L} \dots \textcircled{1}$$

Comparing equation ① with equation

$$\frac{di}{dt} + Ri = \mathcal{E}$$



(C = 4)

$$i(t) = e^{-\frac{R_1}{L}t} \int Q e^{\frac{R_1}{L}t} dt + K e^{-\frac{R_1}{L}t}$$

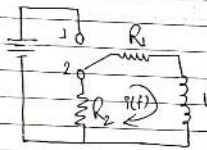
finally general solution is given by

$$i(t) = e^{-\frac{R_1}{L}t} \int \frac{V}{L} e^{\frac{R_1}{L}t} dt + K e^{-\frac{R_1}{L}t}$$

$$= e^{-\frac{R_1}{L}t} \times \frac{V}{L} \times \frac{L}{R_1} e^{\frac{R_1}{L}t} + K e^{-\frac{R_1}{L}t}$$

$$i(t) = \frac{V}{R_1} + K e^{-\frac{R_1}{L}t} \quad \text{--- (1)}$$

At initial condition



$$i(0) = 0$$

Since the inductor opposes the initial current immediately after switching, we get $i(0) = i(0^+) = 0$

Putting value of i in eqn (1)

$$0 = \frac{V}{R_1} + K e^{-\frac{R_1}{L} \times 0}$$

$$K = -\frac{V}{R_1}$$

Now, eqn (1) becomes

$$i(t) = \frac{V}{R_1} - \frac{V}{R_1} e^{-\frac{R_1}{L}t}$$

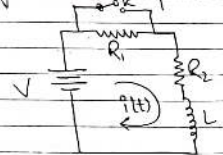
i) Switch K is closed at $t=0$

find $i(t)$ at time $t=0^+$

ii) Switch K is opened at $t=0$. find current response $i(t)$ at $t=0^+$

Solⁿ

When switch is closed



$$V - iR_1 - L \frac{di}{dt} = 0$$

$$L \frac{di}{dt} + R_1 i = V$$

$$\frac{di}{dt} + \left(\frac{R_1}{L}\right)i = \frac{V}{L} \quad \text{--- (1)}$$

Comparing eqn (1) with eqⁿ

$$\frac{di}{dt} + P_1 i = Q$$

its general solution is

$$i(t) = e^{-\frac{R_1}{L}t} \int \frac{V}{L} e^{\frac{R_1}{L}t} dt + K e^{-\frac{R_1}{L}t}$$

$$i(t) = e^{-\frac{R_1}{L}t} \cdot \frac{V}{L} \cdot \frac{L}{R_1} e^{\frac{R_1}{L}t} + K e^{-\frac{R_1}{L}t}$$

$$i(t) = \frac{V}{R_1} + Ke^{\frac{R_2}{L}t} \quad \text{--- (1)}$$

At initial condition, eqⁿ (1) becomes $t=0$,

$$i(0) = \frac{V}{R_1 + R_2}$$

$$\frac{V}{R_1 + R_2} = \frac{V}{R_1} + Ke^{\frac{R_2}{L} \times 0}$$

$$K = \frac{V}{R_1 + R_2} - \frac{V}{R_1}$$

finally, $i(t) = \frac{V}{R_1} + \left(\frac{V}{R_1 + R_2} - \frac{V}{R_1} \right) e^{\frac{R_2}{L}t}$

2nd case,

using KVL

$$V - iR_1 - iR_2 - L \frac{di}{dt} = 0$$

$$L \frac{di}{dt} + (R_1 + R_2)i = V$$

$$\frac{di}{dt} + \left(\frac{R_1 + R_2}{L} \right) i = \frac{V}{L} \quad \text{--- (2)}$$

Comparing eqⁿ (2) with $\frac{di}{dt} + Pi = Q$

and its general solⁿ is

$$i(t) = e^{-\frac{(R_1 + R_2)}{L}t} + \int \frac{V}{L} e^{\frac{(R_1 + R_2)}{L}t} dt + Ke^{-\frac{(R_1 + R_2)}{L}t}$$

$$i(t) = e^{-\frac{(R_1 + R_2)}{L}t} \cdot \frac{V}{L} \cdot \frac{L}{R_1 + R_2} e^{\frac{(R_1 + R_2)}{L}t} + Ke^{-\frac{(R_1 + R_2)}{L}t}$$

$$i(t) = \frac{V}{R_1 + R_2} + Ke^{-\frac{(R_1 + R_2)}{L}t} \quad \text{--- (3)}$$

At initial condition,

$$i(0) = 0, \quad \frac{V}{R_1 + R_2} + Ke^{-\frac{(R_1 + R_2)}{L} \times 0}$$

$$\frac{V}{R_1 + R_2} = \frac{V}{R_1 + R_2} + K$$

$$K = \frac{V}{R_1 + R_2} - \frac{V}{R_1 + R_2}$$

finally exact solution is,

$$i(t) = \frac{V}{R_1 + R_2} + \left(\frac{V}{R_1 + R_2} - \frac{V}{R_1 + R_2} \right) e^{-\frac{(R_1 + R_2)}{L}t}$$

- i) Switch K is moved from 1 to 2 at $t=0$
Find the current response at $t=0^+$
ii) Switch K is moved from 2 to 1 at $t=0$
Find current $i(t)$ at $t=0$

Solⁿ

1st case
using KVL

$$-i - 3i - \frac{di}{dt} = 0$$

$$\frac{di}{dt} + 4i = 0$$

$$\frac{di}{dt} = -4i$$

$$\Rightarrow i = k e^{-4t} \dots \text{--- (1)}$$

At initial condition

$$2i + 3i = 10$$

$$\text{or } i = 2, t = 0$$

Eqⁿ (1) becomes,

$$2 = k e^0 \Rightarrow k = 2$$

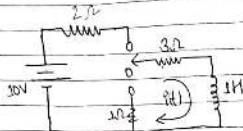
Now, particular solⁿ is

$$i = 2e^{-4t}$$

2nd case,

When switch K is moved from 2 to 1

$$10 - 2i - 3i - \frac{di}{dt} = 0$$



$$\frac{di}{dt} + 5i = 10 \dots \text{--- (1)}$$

Comparing eqⁿ (1) with,

$$\frac{di}{dt} + \gamma i = \delta$$

Its general solution is,

$$i(t) = e^{-\gamma t} \int \delta e^{\gamma t} dt + k e^{-\gamma t}$$

$$i(t) = e^{-5t} \int 10 e^{5t} dt + k e^{-5t}$$

$$i(t) = 2 + k e^{-5t} \dots \text{--- (1)}$$

At initial condition,

$$i(t) = 0, t = 0,$$

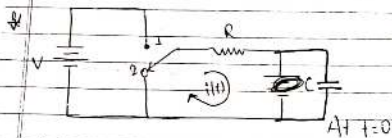
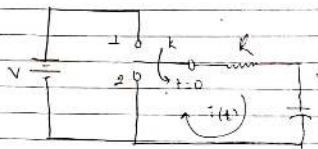
$$k = -2$$

Then particular solution is,

$$i(t) = 2 - 2e^{-5t}$$

Step voltage Response to Simple RC Series circuit

Switch K is moved from position 1 to 2 at time $t=0$ find the charge and current through capacitor at $t=0^+$



Solⁿ

KVL eqⁿ

$$Ri(t) + \frac{1}{C} \int i(t) dt = 0$$

$$R \frac{dQ}{dt} + \frac{Q}{C} = 0$$

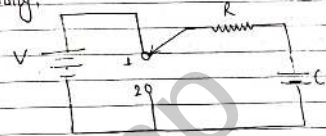
$$\frac{dQ}{dt} + \frac{Q}{RC} = 0$$

$$\frac{dQ}{dt} = -\frac{1}{RC} Q$$

$$\frac{dQ}{Q} = -\frac{1}{RC} dt$$

$$\left[\text{General sol}^n: Q = k e^{-\frac{1}{RC} t} \right]$$

Initially,



Initial charge

$$Q(0^-) = CV$$

Since capacitor does not lose the initial condition current just after switching

$$Q(0^-) = Q(0^+) = CV$$

Put in eqⁿ ① at $t=0^+$

$$CV = k e^{-\frac{1}{RC} \times 0}$$

$$k = CV$$

Put k in ① we get

$$Q(t) = CV e^{-\frac{1}{RC} t}$$

Now current,

$$i(t) = \frac{dQ}{dt}$$

$$= \frac{d}{dt} (CV e^{-\frac{1}{RC} t})$$

$$= CV \frac{d}{dt} e^{-\frac{1}{RC} t} = -CV \times \frac{1}{RC} e^{-\frac{1}{RC} t}$$

$$i(t) = -\frac{V}{R} e^{-\frac{1}{RC} t}$$

particular solⁿ

Apply KVL

$$R i(t) + \frac{1}{C} \int i(t) dt = 0$$

$$\text{Diff} \quad R \frac{di}{dt} + \frac{i}{C} = 0$$

$$\frac{di(t)}{dt} + \frac{i}{RC} = 0$$

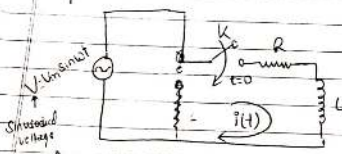
$$\frac{di}{dt} = -\frac{i}{RC}$$

$$\frac{di}{i} = -\frac{1}{RC} dt$$

$$i(t) = k e^{-\frac{t}{RC}}$$

General Soln

If switch K is moved from 1 to 2 at $t=0$. find the current response $i(t)$ at $t=0^+$.



forcing tip

At $t=0$

$$R i(t) + L \frac{di(t)}{dt} = V_m \sin wt$$

$$\frac{di(t)}{dt} + \frac{R}{L} i(t) = \frac{V_m}{L} \sin wt \rightarrow \text{non-homogeneous eqn}$$

Comparing this eqn with

$$\frac{di}{dt} + P i = Q$$

we get $P = \frac{R}{L}$ and $Q = \frac{V_m}{L} \sin wt$

$$\text{So, response } i(t) = e^{-Pt} \int Q e^{Pt} dt + k e^{-Pt}$$

Put P and Q in above eqn and solve

$$i(t) = e^{-\frac{R}{L}t} \int \frac{V_m \sin wt}{L} e^{\frac{R}{L}t} dt + k e^{-\frac{R}{L}t}$$

$$= e^{-\frac{R}{L}t} \frac{V_m}{L} \int \sin wt e^{\frac{R}{L}t} dt + k e^{-\frac{R}{L}t}$$

$$= e^{-\frac{R}{L}t} \frac{V_m}{L} \left[\sin wt \cdot \frac{L}{R} e^{\frac{R}{L}t} - \omega \cos wt \cdot \frac{L^2}{R^2} e^{\frac{R}{L}t} \right] + k e^{-\frac{R}{L}t}$$

$$= e^{-\frac{R}{L}t} e^{\frac{R}{L}t} \frac{V_m}{L} \cdot \frac{L}{R} \left[\sin wt - \omega \cos wt \cdot \frac{L}{R} \right] + k e^{-\frac{R}{L}t}$$

$$i(t) = \frac{V_m}{R} \left[\sin wt - \frac{\omega L}{R} \cos wt \right] + k e^{-\frac{R}{L}t} \quad \text{--- (1)}$$

At initial condition

At $i=0, t=0$

$$0 = \frac{V_m}{R} \left[\sin wt - \frac{\omega L}{R} \cos wt \right] + k$$

$$k = -\frac{V_m}{R} \left(\sin wt - \frac{\omega L}{R} \cos wt \right)$$

Eqⁿ (1) becomes

$$i(t) = \frac{V_m}{R} \left[\sin \omega t - \frac{\omega L}{R} \cos \omega t \right]$$

$$i(t) = \frac{V_m}{R} \left[\sin \omega t - \frac{\omega L}{R} \cos \omega t \right] e^{-\frac{R}{L}t}$$

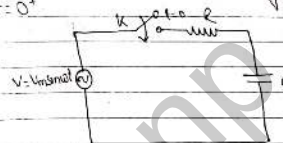
$$i(t) = \frac{V_m}{R} \left[\sin \omega t - \frac{\omega L}{R} \cos \omega t \right] e^{-\frac{R}{L}t}$$

$$K = -\frac{V_m}{R} [0 - \omega L \times 1]$$

$$= \frac{V_m \omega L}{R^2}$$

$$i(t) = \frac{V_m}{R} \left[\sin \omega t - \frac{\omega L}{R} \cos \omega t \right] + \frac{V_m \omega L}{R^2} e^{-\frac{R}{L}t}$$

Switch K is closed at $t=0$. Find the current response at $t=0^+$



KVL eqⁿ is

$$Ri(t) + \frac{1}{C} \int i dt = V_m \sin \omega t \dots (1)$$

Diff eqⁿ (1)

$$R \frac{di}{dt} + \frac{1}{C} i = \omega V_m \cos \omega t$$

$$\frac{di}{dt} + \left(\frac{1}{RC} \right) i = \frac{\omega V_m}{R} \cos \omega t \dots (1)$$

Comparing eqⁿ (1) with

$$\frac{di}{dt} + Pi = Q$$

$$P = \frac{1}{RC}, Q = \frac{\omega V_m}{R} \cos \omega t$$

It's general solⁿ is

$$i(t) = e^{-\frac{1}{RC}t} \int e^{\frac{1}{RC}t} \frac{\omega V_m \cos \omega t}{R} dt + K e^{-\frac{1}{RC}t}$$

$$i(t) = e^{-\frac{1}{RC}t} \frac{\omega V_m}{R} \int \cos \omega t e^{\frac{1}{RC}t} dt + K e^{-\frac{1}{RC}t}$$

Taking $I = \int \cos \omega t e^{-t/\tau} dt$

$$= \cos \omega t \int e^{-t/\tau} dt - \int \left(\frac{d \cos \omega t}{dt} \int e^{-t/\tau} dt \right) dt$$

$$= RC \cos \omega t e^{-t/\tau} - \int (-R\omega \sin \omega t e^{-t/\tau}) dt$$

$$= RC \cos \omega t e^{-t/\tau} + R\omega C \int \sin \omega t e^{-t/\tau} dt$$

$$= RC \cos \omega t e^{-t/\tau} + R\omega C \left[\sin \omega t \int e^{-t/\tau} dt - \int \left(\frac{d \sin \omega t}{dt} \int e^{-t/\tau} dt \right) dt \right]$$

$$= RC \cos \omega t e^{-t/\tau} + R\omega C \left[RC \sin \omega t e^{-t/\tau} - \omega RC \int \cos \omega t e^{-t/\tau} dt \right]$$

$$I = RC \cos \omega t e^{-t/\tau} + R^2 C^2 \omega \sin \omega t e^{-t/\tau} - R^2 C^2 \omega^2 \int \cos \omega t e^{-t/\tau} dt$$

$$I + R^2 C^2 \omega^2 \int \cos \omega t e^{-t/\tau} dt = RC \cos \omega t e^{-t/\tau} + R^2 C^2 \omega \sin \omega t e^{-t/\tau}$$

$$I = \frac{1}{1 + R^2 C^2 \omega^2} \left(RC \cos \omega t e^{-t/\tau} + R^2 C^2 \omega \sin \omega t e^{-t/\tau} \right)$$

$$\therefore i(t) = e^{-t/\tau} \cdot \frac{\omega V_m}{R} \cdot \frac{1}{1 + R^2 C^2 \omega^2} \left[RC \cos \omega t + R^2 C^2 \omega \sin \omega t \right] + k e^{-t/\tau}$$

$$= \frac{\omega V_m C}{1 + R^2 C^2 \omega^2} \left[\cos \omega t + RC \omega \sin \omega t \right] + k e^{-t/\tau} \quad \text{--- (ii)}$$

At initial condition,

$$i(t) = 0, t = 0$$

$$0 = \frac{\omega V_m C}{1 + R^2 C^2 \omega^2} [1] + k e^0$$

$$k = - \frac{\omega V_m C}{1 + R^2 C^2 \omega^2}$$

finally eqn (ii) becomes,

$$i(t) = \frac{\omega V_m C}{1 + R^2 C^2 \omega^2} (\cos \omega t + RC \omega \sin \omega t) - \frac{\omega V_m C}{1 + R^2 C^2 \omega^2} e^{-t/\tau} \quad \text{--- (iii)}$$

Time Constant (τ)

Significance

Simple RL circuit

$$i = \frac{V}{R} e^{-\frac{R}{L}t} \quad \text{--- (i)}$$

Simple RC circuit

$$i = \frac{V}{R} e^{-t/\tau} \quad \text{--- (ii)}$$

$$\text{let } \frac{V}{R} = i_0$$

Then equation becomes,

$$i = i_0 e^{-\frac{t}{\tau}}$$

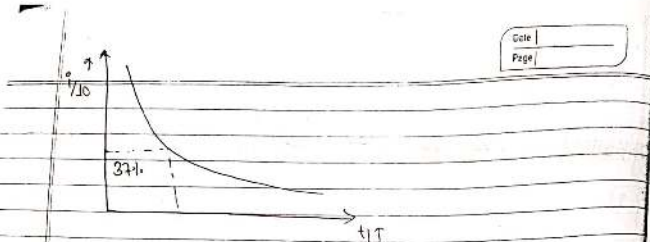
$$\frac{i}{i_0} = e^{-t/\tau}$$

where (τ = time constant = L/R)

$$\text{At } t = \tau$$

$$\frac{i}{i_0} = e^{-1} = \frac{1}{e} = 37\%$$

The current which is decreased by 63.1% of initial current



Calculation of initial values and their derivatives.

Characteristics of various elements

Resistors: The initial value of resistor changes instantaneously just after switching.

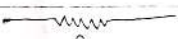
Inductors: The initial value of inductors does not change instantaneously just after switching. It means the inductor act as open circuit at time $t=0^+$. But if inductors have initial current, it will act as the current source at $t=0^+$.

Capacitors:

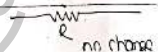
The initial value of capacitors does not change instantaneously just after switching. That means it act as short circuit at time $t=0^+$. But if it has initial voltage it will act as voltage source at $t=0^+$.

Summary

Elements at $t=0$



Elements at $t=0^+$



Source

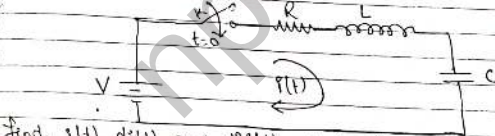
switch

o/c

short circuit

$V = q/c$

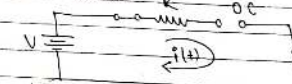
Switch k is closed at $t=0$



find $i(t)$, $\frac{di(t)}{dt}$ and $\frac{d^2i(t)}{dt^2}$ at $t=0^+$

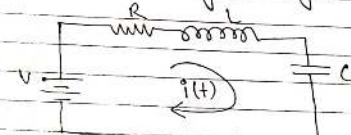
At $t=0^+$

Equivalent circuit



since circuit is open sc, current is zero

To find $\frac{di(t)}{dt}$ and $\frac{d^2i(t)}{dt^2}$ we need to write the general equation from the given circuit



KVL eqn

$$Ri(t) + L\frac{di(t)}{dt} + \frac{1}{C}\int i(t)dt = V \dots (1)$$

Note: If current is given then s.c & o.c are used in place of capacitor & inductor respectively.

Date /
Page

Since the circuit is open $\frac{1}{C} \int i(t) dt = 0$

At $t=0^+$

$$R \times 0 + L \frac{di(t)}{dt} = V$$

$$\frac{di(t)}{dt} = \frac{V}{L} \text{ Als}$$

Differentiating eqn ①

$$R \frac{di(t)}{dt} + L \frac{d^2i(t)}{dt^2} + \frac{i(t)}{C} = 0$$

$$\text{At } t=0^+, \frac{di(t)}{dt} = \frac{V}{L}, i(t)=0$$

$$R \frac{V}{L} + L \frac{d^2i(t)}{dt^2} + \frac{0}{C} = 0$$

$$\frac{d^2i(t)}{dt^2} = -\frac{VR}{L^2} \text{ Als}^2$$

(A)

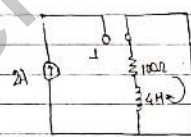
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Page

In the circuit shown, the switch is moved from 1 to 2 at $t=0$. Find $I(t)$



Solution

At $t=0$



$$100I(t) + 4 \frac{dI(t)}{dt} = 0$$

$$\text{or, } \frac{dI(t)}{dt} + \frac{100}{4} I(t) = 0$$

$$\text{or, } \frac{dI(t)}{dt} = -25 I(t)$$

$$\Rightarrow \frac{dI(t)}{I(t)} = -25 dt$$

$$\text{General solution } I(t) = k e^{-25t} \dots \text{①}$$

$$\left[-2 \text{ kA} = -2e^0 \right]$$

Initial condition

$$I(0) = -2$$

Put in eqn ①

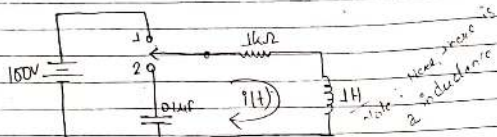
$$-2 = k e^{-25 \times 0}$$

$$\Rightarrow k = -2$$

Hence,

$$I(t) = -2 e^{-25t}$$

In the circuit shown, the switch is moved from 1 to 2 at $t=0$. find $i(t)$, $\frac{di(t)}{dt}$, $\frac{d^2i(t)}{dt^2}$ at $t=0^+$.



Solution.

At $t=0^+$, the inductor carries same current after switching which is before switching SD.

At 1 Initially, $i(0^-) = \frac{100}{1k\Omega} = \frac{100}{1 \times 10^3} = 0.1A$

for inductor $i(0^-) = i(0^+) = 0.1A$

Now, to find $\frac{di(t)}{dt}$ and $\frac{d^2i(t)}{dt^2}$ we go to position 2

$$Ri(t) + L \frac{di(t)}{dt} + \frac{1}{C} \int i(t) dt = 0 \dots \dots \textcircled{1}$$

voltage across capacitance

At $t=0^+$

$$R \times 0.1 + L \frac{di(t)}{dt} + 0 = 0$$

$$\text{or, } \frac{di(t)}{dt} = \frac{-1000 \times 0.1}{1} \text{ A/s}$$

Diff eqn $\textcircled{1}$ we get

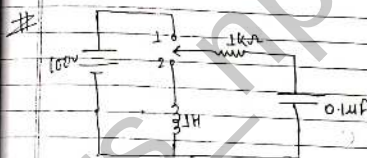
$$R \frac{di(t)}{dt} + L \frac{d^2i(t)}{dt^2} + \frac{1}{C} i(t) = 0$$

find initial condition
initial to find + 20ms

$$(1000)(-100) + 1^2 \frac{d^2i(t)}{dt^2} + \frac{0.1}{0.1 \times 10^{-6}} = 0$$

$$\text{or, } \frac{d^2i(t)}{dt^2} = \frac{10^5 - 10^6}{1} = -9 \times 10^5 \checkmark$$

$\frac{10^5 - 10^6}{1} = \frac{10^5(1 - 10)}{1} = -9 \times 10^5$



Soln

At $t=0$

At 1 initially $i(0^-) = \frac{100}{1k\Omega} = 0.1A$

for capacitance $i(0^-) = i(0^+) = 0.1A$

Now, to find $\frac{di(t)}{dt}$ and $\frac{d^2i(t)}{dt^2}$ we go to position 2

$$Ri(t) + L \frac{di(t)}{dt} + \frac{1}{C} \int i(t) dt = 0 \dots \dots \textcircled{1}$$

$$\text{or, } 1000 \times 0.1 + 1 \frac{di}{dt} + 100 = 0$$

$$\frac{di(t)}{dt} = -20 \text{ A/s}$$

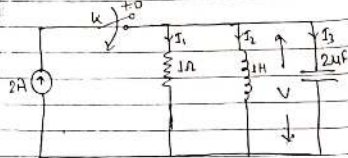
Diff eqn $\textcircled{1}$

$$\frac{d^2i(t)}{dt^2} + R \frac{di(t)}{dt} + \frac{1}{C} i(t) = 0$$

$$\frac{d^2i(t)}{dt^2} + \frac{20}{1 \times 10^{-3}} + \frac{0.1}{0.1 \times 10^{-6}} = 0 \Rightarrow \frac{d^2i(t)}{dt^2} = \frac{-10^6 \text{ A/s}^2}{9.8 \times 10^5 \text{ A/s}^2}$$

Parallel Circuit

eg. switch k is closed at $t=0$. find the voltage $v(t)$, $\frac{dv(t)}{dt}$, $\frac{d^2v(t)}{dt^2}$ at $t=0^+$.



Solⁿ

At $t=0$, (kcl eqⁿ)

$$v(0^-) = v(0^+) = 0$$

To find $\frac{dv(t)}{dt}$, $\frac{d^2v(t)}{dt^2}$ we write the general eqⁿ at $t=0$

$$2 - I_1 - I_2 - I_3 = 0$$

$$I_1 = \frac{v}{R}, \quad I_2 = \frac{1}{L} \int v dt, \quad I_3 = C \frac{dv}{dt}$$

$$\Rightarrow 2 = \frac{v}{R} + \frac{1}{L} \int v dt + C \frac{dv}{dt} \quad \text{--- (1)}$$

At $t=0^+$

$$2 = \frac{0}{R} + 0 + 2\mu F \frac{dv}{dt}$$

$$\text{Or, } \frac{dv}{dt} = \frac{2}{2\mu F}$$

Diff. eqⁿ (1)

$$0 = \frac{1}{R} \frac{dv}{dt} + \frac{v}{L} + C \frac{d^2v}{dt^2}$$

Put values of v , $\frac{dv}{dt}$ at $t=0$ then

$$C \frac{d^2v}{dt^2} = 0 + \frac{2}{2 \times 10^{-6}}$$

$$\Rightarrow \frac{d^2v}{dt^2} = -10^6$$

$$\text{Or, } \frac{d^2v}{dt^2} = \frac{10^6}{2 \times 10^{-6}} = \frac{10^{12}}{2} = -5 \times 10^{11} \text{ Vs}^{-2}$$

Important formula

Capacitor

$$I_C = C \frac{dv}{dt}$$

voltage

$$V_C = \frac{1}{C} \int i dt$$

Inductor

$$I_L = \frac{1}{L} \int v dt$$

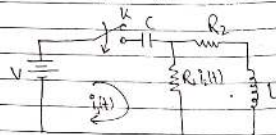
voltage

$$V_L = L \frac{di}{dt}$$

* Two mesh example

The circuit is initially relaxed and switch k is closed at $t=0$. Find $i_1(t)$, $i_2(t)$, $\frac{di_1(t)}{dt}$, $\frac{di_2(t)}{dt}$, $\frac{d^2i_1(t)}{dt^2}$ and $\frac{d^2i_2(t)}{dt^2}$

Solution.



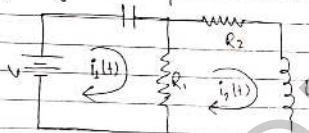
At $t=0^+$, equivalent circuit is



$$i_1(0^+) = \frac{V}{R_1}, \quad i_2(0^+) = 0$$

$$i_1(t) = \frac{V}{R_1}, \quad i_2(t) = 0$$

To find of other requirements, we write general eqⁿ at $t=0$



from first mesh using KVL

$$\frac{1}{C} \int i_1(t) dt + R_1 [i_1(t) - i_2(t)] - V = 0 \quad \text{--- (i)}$$

from 2nd mesh using KVL

$$R_2 i_2(t) + L \frac{di_2(t)}{dt} + R_1 [i_2(t) - i_1(t)] = 0 \quad \text{--- (ii)}$$

from eqⁿ (i)

$$\text{At } t=0^+ \quad R_1 x_0 + L \frac{di_2(t)}{dt} + R_1 (0 - \frac{V}{R_1}) = 0$$

$$L \frac{di_2(t)}{dt} = V$$

$$\frac{di_2(t)}{dt} = \frac{V}{L}$$

Diff. eqⁿ (ii) w.r.t 't'

$$\frac{i_1(t)}{C} + R_1 \frac{di_1(t)}{dt} - R_1 \frac{di_2(t)}{dt} = 0 \quad \text{--- (iii)}$$

At $t=0^+$

$$\frac{V}{R_1 C} + R_1 \frac{di_1(t)}{dt} - R_1 \frac{V}{L} = 0$$

$$R_1 \frac{di_1(t)}{dt} = R_1 \frac{V}{L} - \frac{V}{R_1 C}$$

$$\frac{di_1(t)}{dt} = \frac{V}{L} - \frac{V}{R_1^2 C}$$

Again, Diff eqn ①

$$R_2 \frac{di_1(t)}{dt} + L \frac{d^2 i_1(t)}{dt^2} + R_1 \frac{di_1(t)}{dt} - R_1 \frac{di_2(t)}{dt} = 0$$

$$0, R_2 \frac{V}{L} + L \frac{d^2 i_1(t)}{dt^2} + R_1 \frac{V}{L} - R_1 \left(\frac{V}{L} - \frac{V}{R_1 C} \right) = 0$$

$$0, R_2 \frac{V}{L} + L \frac{d^2 i_1(t)}{dt^2} + R_1 \frac{V}{L} - \frac{R_1 V}{L} + \frac{V}{R_1 C} = 0$$

$$\frac{d^2 i_1(t)}{dt^2} = -\frac{V}{R_1 C L} - \frac{R_2 V}{L^2}$$

Also,

Diff eqn ② w.r.t. t

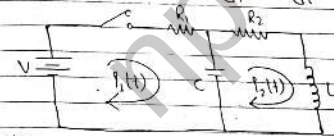
$$\frac{1}{C} \frac{di_1(t)}{dt} + R_1 \frac{d^2 i_1(t)}{dt^2} - R_1 \frac{d^2 i_2(t)}{dt^2} = 0$$

$$0, \frac{1}{C} \left(\frac{AV}{L} - \frac{V}{R_1 C} \right) + R_1 \frac{d^2 i_1(t)}{dt^2} - R_1 \left(-\frac{V}{R_1 C L} - \frac{R_2 V}{L^2} \right) = 0$$

$$0, \frac{V}{CL} - \frac{V}{R_1^2 C^2} + R_1 \frac{d^2 i_1(t)}{dt^2} + \frac{V}{CL} + \frac{R_1 R_2 V}{L^2} = 0$$

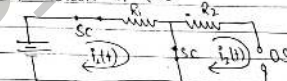
$$\frac{d^2 i_1(t)}{dt^2} = \frac{V}{R_1^3 C^2} - \frac{2V}{R_1 CL} - \frac{R_2 V}{L^2}$$

The circuit is initially relaxed. The switch K is closed at $t=0$. Find $i_1(t)$, $i_2(t)$, $\frac{d^2 i_1(t)}{dt^2}$, $\frac{d^2 i_2(t)}{dt^2}$, $\frac{di_1(t)}{dt}$ and $\frac{di_2(t)}{dt}$



Solution,

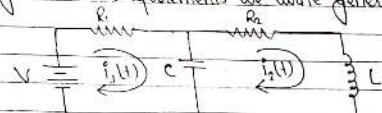
Equivalent circuit at $t=0^+$



$$i_1(0^+) = \frac{V}{R_1} \text{ and } i_2(0^+) = 0$$

Now,

To find other requirements we write general eqn at $t=0$



1st mesh,

$$R_1 i_1(t) + \frac{1}{C} \int [i_1(t) - i_2(t)] dt = V \quad \text{--- (1)}$$

2nd mesh,

$$R_2 i_2(t) + L \frac{di_2(t)}{dt} + \frac{1}{C} \int [i_2(t) - i_1(t)] dt = 0 \quad \text{--- (2)}$$

Diff. eqn ① w.r.t 't'

$$R_1 \frac{di_1(t)}{dt} + \frac{i_1 - i_2}{C} = 0$$

At $t=0^+$

$$R_1 \frac{di_1(t)}{dt} + \frac{V - 0}{RC} = 0$$

$$\frac{di_1(t)}{dt} = -\frac{V}{R_1 C}$$

From eqn ①

Total voltage across capacitance = 0

$$R_2 i_2(t) + L \frac{di_2(t)}{dt} + 0 = 0$$

$$R_2 \times 0 + L \frac{di_2(t)}{dt} + 0 = 0$$

$$\frac{di_2(t)}{dt} = 0$$

Again,

Diff. eqn ① w.r.t

$$R_1 \frac{d^2 i_1(t)}{dt^2} + \frac{1}{C} \frac{di_1(t)}{dt} - \frac{1}{C} \frac{di_2(t)}{dt} = 0$$

$$R_1 \frac{d^2 i_1(t)}{dt^2} + \frac{1}{C} \left(-\frac{V}{R_1 C} \right) = \frac{1}{C} \times 0 = 0$$

$$\frac{d^2 i_1(t)}{dt^2} = \frac{V}{R_1^2 C}$$

Again,

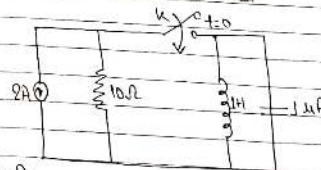
Differentiating eqn ①

$$R_2 \frac{di_2(t)}{dt} + L \frac{d^2 i_2(t)}{dt^2} + \frac{i_2(t)}{C} - \frac{i_1(t)}{C} = 0$$

$$R_2 \times 0 + L \frac{di_2(t)}{dt} + 0 = \frac{V}{R_1 C} = 0$$

$$\frac{d^2 i_2(t)}{dt^2} = \frac{V}{R_1 C L}$$

* The circuit is initially relaxed. The switch k is closed at $t=0$. Find $v(t)$, $di_1(t)$ and $d^2 v$ at $t=0^+$



At $t=0$,

$$V(0^+) = V(0^-) = 10 \times 2 = 20V \checkmark$$

Then

When switch is closed

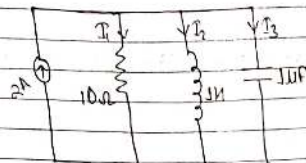
$$2 = I_1 + I_2 + I_3 \quad (1)$$

At $t=0$, then

$$I_1 = \frac{V(t)}{10}$$

$$I_2 = \frac{1}{L} \int V(t) dt$$

$$I_3 = C \frac{dv}{dt}$$



Then eqn ① becomes,

$$2 = \frac{V(t)}{10} + \frac{1}{L} \int V(t) dt + C \frac{dv(t)}{dt} \quad \text{--- ①}$$

Initially no current across inductor,

$$2 = \frac{V(t)}{10} + 0 + C \frac{dv(t)}{dt}$$

$$2 = \frac{20}{10} + 0 + C \frac{dv(t)}{dt}$$

$$\frac{dv(t)}{dt} = 0 \text{ V/s}$$

Diff. eqn ① w.r.t. (t) ,

$$0 = \frac{1}{10} \frac{dv(t)}{dt} + \frac{V(t)}{L} + C \frac{d^2V(t)}{dt^2}$$

$$0 = \frac{1}{10} \times 0 + \frac{20}{10} + C \frac{d^2V(t)}{dt^2}$$

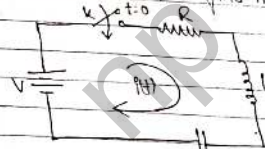
$$\frac{d^2V(t)}{dt^2} = \frac{-20}{10^{-6}}$$

$$\frac{d^2V(t)}{dt^2} = -2 \times 10^7 \text{ V/s}^2$$

Second Order Equation

RLC Series circuit

Switch K is closed at $t=0$ find the current response.



Soln

At $t=0$, using KVL

$$R i(t) + L \frac{di(t)}{dt} + \frac{1}{C} \int i(t) dt = V \quad \text{--- ①}$$

Differentiating eqn ① w.r.t. t

$$R \frac{di(t)}{dt} + L \frac{d^2i(t)}{dt^2} + \frac{i(t)}{C} = 0$$

$$\frac{d^2i(t)}{dt^2} + \frac{R}{L} \frac{di(t)}{dt} + \frac{i(t)}{LC} = 0$$

Characteristics eqn

$$m^2 + \left(\frac{R}{L}\right)m + \left(\frac{1}{LC}\right) = 0$$

Soln of this equation. This eqn has two roots say s_1 & s_2 .

General soln is

$$i(t) = k_1 e^{s_1 t} + k_2 e^{s_2 t} \quad \text{--- ②}$$

The quadratic eqn has solution

$$= \frac{-R}{2L} \pm \frac{1}{2L} \sqrt{R^2 - \frac{4L}{C}}$$

If $R^2 > 4L/C$

The roots will be real, distinct and positive

$$i(t) = k_1 e^{s_1 t} + k_2 e^{s_2 t}$$

If $R^2 = \frac{4L}{C}$

The roots will be repeated.

$$i(t) = (k_1 + k_2 t) e^{s t}$$

If $R^2 < \frac{4L}{C}$

The roots will be imaginary
let,

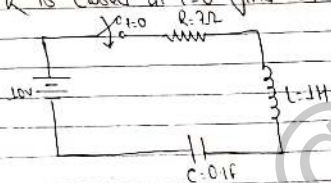
$$s_1 = \alpha - j\beta$$

$$s_2 = \alpha + j\beta$$

So, its general solⁿ is

$$i(t) = k_1 \cos \beta t + k_2 \sin \beta t$$

Switch k is closed at $t=0$ find $i(t)$



Solution.

At $t=0$,

KVL eqⁿ is

$$R \frac{di(t)}{dt} + L \frac{d^2 i(t)}{dt^2} + \frac{1}{C} \int i(t) dt = V \dots \textcircled{1}$$

Diff. eqⁿ ① w.r.t t

$$\frac{d^2 i(t)}{dt^2} + 7 \frac{di(t)}{dt} + 10 i(t) = 0 \dots \textcircled{2}$$

It's characteristics eqⁿ is

$$m^2 + 7m + 10 = 0$$

$$\alpha, m = -5, -2$$

The general solution is

$$i(t) = k_1 e^{s_1 t} + k_2 e^{s_2 t} \dots \textcircled{3}$$

Initially,

$$i(0^-) = i(0^+) = 0$$

Eqⁿ ③ becomes,

$$0 = k_1 e^{-5 \times 0} + k_2 e^{-2 \times 0}$$

$$\Rightarrow k_1 + k_2 = 0 \dots \textcircled{4}$$

Diff. eqⁿ ③

$$\frac{di(t)}{dt} = -5k_1 e^{-5t} - 2k_2 e^{-2t} \dots \textcircled{5}$$

From eqⁿ ④, at $t=0$,

$$7 \times 0 + \frac{di(t)}{dt} + 0 = 10 \Rightarrow \frac{di(t)}{dt} = 10$$

$$\frac{di(t)}{dt} = 10$$

Eqn (ii) becomes,
 $-5k_1 - 2k_2 = 10 \dots (*)$

Solving eqn (i) and (ii)

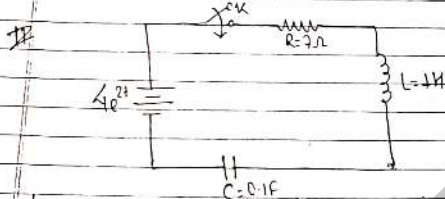
$$k_2 = 10/3 \text{ \& } k_1 = -10/3$$

Finally particular soln is

$$i(t) = -10/3 e^{-5t} + 10/3 e^{-2t}$$

$$i(t) = 10/3 e^{-2t} - 10/3 e^{-5t}$$

which is exact solution



In above question voltage 10V is replaced by $4e^{2t}$.

Soln

At $t=0$

KVL eqn is

$$Ri(t) + L \frac{di(t)}{dt} + \frac{1}{C} \int i(t) dt = 4e^{2t} \dots (1)$$

Diff eqn (i) w.r.t to t

$$\frac{d^2 i(t)}{dt^2} + 7 \frac{di(t)}{dt} + 10i(t) = 8e^{2t} \dots (ii)$$

It

which is non-homogeneous linear differential eqn of 2nd order.

So it has two solution

i.e. homogeneous and particular

for homogeneous solution $i(t)_h$

characteristic eqn is,

$$m^2 + 7m + 10 = 0$$

$$\Rightarrow m = -2, -5$$

$$i(t)_h = k_1 e^{-5t} + k_2 e^{-2t} \dots (iii)$$

for particular soln $i(t)_p$

$$i(t)_p = ce^{2t}$$

$$i(t)_p' = 2ce^{2t}$$

$$i(t)_p'' = 4ce^{2t}$$

Substituting these value in eqn (i)

$$4ce^{2t} + 14ce^{2t} + 10ce^{2t} = 8e^{2t}$$

$$a_1, 28ce^{2t} = 8e^{2t}$$

$$a_1, c = \frac{8}{28} = \frac{2}{7}$$

Finally General Solution is,

$$i(t) = k_1 e^{-5t} + k_2 e^{-2t} + \frac{2}{7} e^{2t} \dots (iv)$$

At $t=0$, $i(t)=0$ so,

$$0 = k_1 + k_2 + \frac{2}{7}$$

$$\Rightarrow k_1 + k_2 = -\frac{2}{7} \dots (v)$$

Differentiating eqn (iv) w.r.t 't'.

$$\frac{di(t)}{dt} = -5k_1 e^{-5t} - 2k_2 e^{-2t} + \frac{4}{7} e^{2t} \dots (vi)$$

From eqn (i)

Initial condition $t=0$

$$7 \times 0 + \frac{di(t)}{dt} + 0 = 4e^{2t}$$

Eqn (vi) becomes,

$$4e^{2t} = -5k_1 e^{-5t} - 2k_2 e^{-2t} + \frac{4}{7} e^{2t}$$

At $t=0$,

$$4 = -5k_1 - 2k_2 + \frac{4}{7}$$

$$a_1, -5k_1 - 2k_2 = \frac{24}{7} \dots (vii)$$

Solving eqn (v) and (vii)

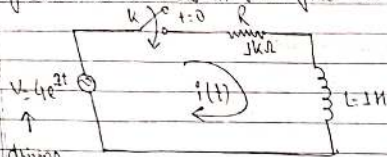
$$k_2 = \frac{2}{3}, k_1 = -\frac{2}{3}$$

$$\therefore i(t) = \frac{2}{3} e^{-5t} + \frac{2}{3} e^{-2t} + \frac{2}{7} e^{2t}$$

$$\therefore i(t) = \frac{2}{3} e^{-5t} + \frac{2}{3} e^{-2t} + \frac{2}{7} e^{2t}$$

RL circuit with driving force

eg switch K is closed at $t=0$. find the current response



driving force 801^u

At $t=0$, using KVL

$$Ri(t) + L \frac{di(t)}{dt} = 4e^{-2t} \dots \textcircled{1}$$

$$\frac{di(t)}{dt} + \frac{R}{L}i(t) = \frac{4}{L}e^{-2t}$$

characteristics eqⁿ

$$m + \frac{R}{L} = 0 \Rightarrow m = -\frac{R}{L}$$

complementary eqⁿ

$$i(t) = ke^{-\frac{R}{L}t}$$

for particular solⁿ, let trial solⁿ as $i_p = ke^{-2t}$

$$\frac{di_p}{dt} = -2ke^{-2t}$$

Put values in eqⁿ ①

$$-2ke^{-2t} + R \times ke^{-2t} = 4e^{-2t}$$

$$-2k + 1000k = 4 \quad \text{or } 998k = 4$$

$$\Rightarrow k = \frac{4}{998}$$

Now, the total general solution is

$i(t) = \text{complementary function} + \text{particular integral}$
 $i(t) = ke^{-\frac{R}{L}t} + \frac{4}{998}e^{-2t}$

Initially, the current through the inductor is zero which is also the value after switching so,

$$i(0) = i(0^+) = 0$$

Put in general eqⁿ

$$0 = ke^{-\frac{R}{L} \cdot 0} + \frac{4}{998}e^{-2 \cdot 0}$$

$$\Rightarrow k = -\frac{4}{998}$$

The time response is

$$i(t) = -\frac{4}{998}e^{-\frac{R}{L}t} + \frac{4}{998}e^{-2t}$$

OR

comparing eqⁿ ① with $\frac{di}{dt} + Pi = Q \Rightarrow P = 1000, Q = 4e^{-2t}$

$$i(t) = e^{-1000t} \int \frac{4e^{-2t}}{e^{-1000t}} dt + \frac{ke^{-1000t}}{1000-2}$$

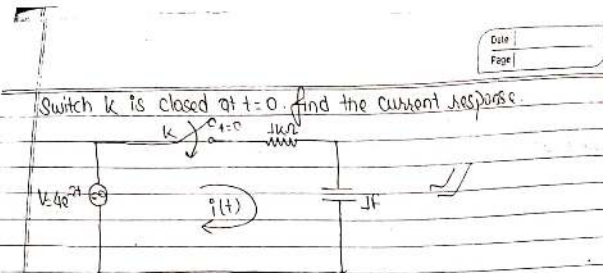
$$= e^{-1000t} \left[\frac{4}{998} e^{998t} \right] + \frac{ke^{-1000t}}{998} = \frac{4e^{-2t}}{998} + \frac{ke^{-1000t}}{998}$$

At initial condition, $i(0^+) - i(0^-) = 0$

$$0 = \frac{4}{998} + k \Rightarrow k = -\frac{4}{998}$$

finally exact solⁿ is

$$i(t) = \frac{4}{998}e^{-2t} - \frac{4}{998}e^{-1000t}$$



Solⁿ

At $t=0^-$ KVL eqn is

$$Ri(t) + \frac{1}{C} \int i(t) dt = 4e^{-2t}$$

$$1000i(t) + \int i(t) dt = 4e^{-2t} \quad \text{--- (1)}$$

Diff w.r.t. 't'

$$1000 \frac{di(t)}{dt} + i(t) = -8e^{-2t}$$

$$\frac{di(t)}{dt} + \frac{1}{1000} i(t) = \left(\frac{-8}{1000} \right) e^{-2t} = \frac{-1}{125} e^{-2t}$$

Comparing it with $\frac{dy}{dt} + Py = Q$

$$P = \frac{1}{1000}, Q = \frac{-1}{125}$$

$$i(t) = e^{-\frac{t}{1000}} \int e^{\frac{t}{1000}} \left(\frac{-1}{125} e^{-2t} \right) dt + k e^{\frac{t}{1000}}$$

$$= \frac{-1}{125} e^{-\frac{t}{1000}} \int e^{\frac{t}{1000} - 2t} dt + k e^{-\frac{t}{1000}}$$

$$= \frac{-1}{125} e^{-\frac{t}{1000}} \cdot \frac{e^{\frac{t}{1000} - 2t}}{\frac{1}{1000} - 2} + k e^{-\frac{t}{1000}}$$

$$= \frac{8}{1999} e^{-2t} + k e^{-\frac{t}{1000}}$$

At $t=0, i(t)=0$

$$0 = \frac{8}{1999} + k \Rightarrow k = -\frac{8}{1999}$$

$$i(t) = \frac{8}{1999} e^{-2t} - \frac{8}{1999} e^{-\frac{t}{1000}} \quad \text{--- (2)}$$

$$Ri(0^+) = 4e^{-2t}$$

$$i(0^+) = \frac{4e^{-2t}}{R}$$

Put in total Solⁿ at $t=0$

$$4e^{-2t} = k e^{-\frac{t}{1000}} - \frac{8}{1999} e^{-2t}$$

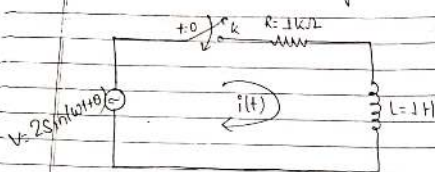
$$\frac{4}{R} = \frac{k - 8}{1 - 2RC}$$

$$\therefore k = \frac{4}{R} + \frac{8}{1 - 2RC}$$

Put R in total Solⁿ

$$i(t) = \frac{8}{1999} e^{-2t} + \left(\frac{1}{125} - \frac{8}{1999} \right) e^{-\frac{t}{1000}}$$

Switch k is closed at $t=0$. find current response $i(t)$



Solution

At $t=0$, using KVL

$$Ri(t) + L \frac{di(t)}{dt} = 2 \sin(wt + \theta)$$

$$\frac{di(t)}{dt} + \frac{R}{L} i(t) = \frac{2}{L} \sin(wt + \theta)$$

$$\frac{di(t)}{dt} + i(t) = 2 \sin(wt + \theta) \dots \text{--- (1)}$$

Now characteristics eqⁿ is

$$m+1=0$$

$$m=-1$$

So

$$i(t)_h = ke^{-t} \dots \text{--- (ii) complementary function}$$

for particular solⁿ

$$i_p = A \cos(wt + \theta) + B \sin(wt + \theta) \dots \text{--- (iii)}$$

$$i(t)_p = A \cos(wt + \theta) + B \sin(wt + \theta)$$

Substituting these value in eqⁿ (1)

$$-A \omega \sin(wt + \theta) + B \omega \cos(wt + \theta) + A \cos(wt + \theta) + B \sin(wt + \theta) = 2 \sin(wt + \theta)$$

$$-A \omega + B = 2$$

$$B \omega + A = 0$$

By eqⁿ

Solving these

$$-A \omega + B = 2$$

$$B \omega + A = 0$$

$$B(\omega^2 + 1) = 2$$

$$\Rightarrow B = \frac{2}{\omega^2 + 1}$$

$$\left(\frac{2}{\omega^2 + 1}\right) \omega + A = 0$$

$$A = -\frac{2\omega}{\omega^2 + 1}$$

Substituting these value in (iii), we get

$$i(t) = \frac{-2\omega \cos(wt + \theta)}{\omega^2 + 1} + \frac{2 \sin(wt + \theta)}{\omega^2 + 1}$$

finally,

$$i(t) = ke^{-t} \frac{2\omega \cos(wt + \theta)}{\omega^2 + 1} + \frac{2 \sin(wt + \theta)}{\omega^2 + 1}$$

At initial $t=0$, $i(t)=0$

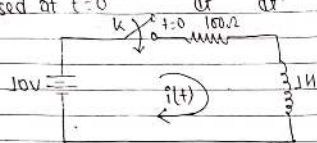
$$0 = ke^0 \frac{2\omega \cos(wt + \theta)}{\omega^2 + 1} + \frac{2 \sin(wt + \theta)}{\omega^2 + 1}$$

$$k = \frac{2\omega \cos(wt + \theta)}{\omega^2 + 1} - \frac{2 \sin(wt + \theta)}{\omega^2 + 1}$$

$$\text{finally, } i(t) = \frac{2}{\omega^2 + 1} [\omega \cos(wt + \theta) - \sin(wt + \theta)] e^{-t} = \frac{2\omega \cos(wt + \theta)}{\omega^2 + 1} - \frac{2 \sin(wt + \theta)}{\omega^2 + 1}$$

(A)

Find the current $i(t)$, $\frac{di(t)}{dt}$, $\frac{d^2i(t)}{dt^2}$ at $t=0^+$ switch k is closed at $t=0$



Solution.

Since ckt is open before switching the initial current is $i(0^-) = 0$
by nature of inductor, it does carry initial current just after switching $i(0^+) = i(0^-) = 0$ A

Now,

At $t=0$, switch is closed and general eqn is

$$100i + 1 \frac{di(t)}{dt} = 10 \quad \text{--- (1)}$$

At $t=0^+$

$$100 \times 0 + \frac{di(t)}{dt} = 10$$

$$\frac{di(t)}{dt} = 10 \text{ A/s}^{-1}$$

Diff eqn (1), we get

$$100 \frac{di(t)}{dt} + \frac{d^2i(t)}{dt^2} = 0$$

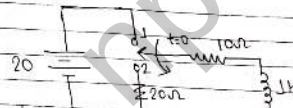
$$\text{At } t=0^+, \frac{d^2i(t)}{dt^2} = -100 \frac{di(t)}{dt}$$

$$\frac{d^2i(t)}{dt^2} = -100 \times 10 = -1000 \text{ A/s}^{-2}$$

Date
Page

k is moved from 1 to 2 at $t=0$ steady state condition having reached before switching find $i(t)$, $\frac{di(t)}{dt}$ and $\frac{d^2i(t)}{dt^2}$ at $t=0^+$

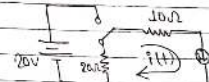
Soln



Since ckt is closed to 1

$$i(0^-) = \frac{20}{20} = 2 \text{ A}$$

$$i(0^+) = 2 \text{ A}$$



Now,

At $t=0$ switch is closed to 2.

General eqn is

$$20i(t) + 10i(t) + \frac{di(t)}{dt} = 0 \quad \text{--- (1)}$$

$$\text{Or, } \frac{di(t)}{dt} + 30i(t) = 0$$

$$\text{At } t=0^+, \frac{di(t)}{dt} = -30 \times 2 = -60 \text{ A/s}^{-1}$$

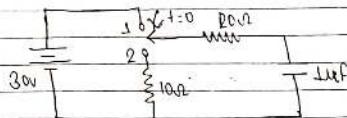
$$\text{Diff eqn (1)} \quad 30 \frac{di(t)}{dt} + \frac{d^2i(t)}{dt^2} = 0$$

$$\text{At } t=0^+ \quad \frac{d^2i(t)}{dt^2} = -30 \times (-60)$$

$$\Rightarrow \frac{d^2i(t)}{dt^2} = 1800 \text{ A/s}^{-2}$$

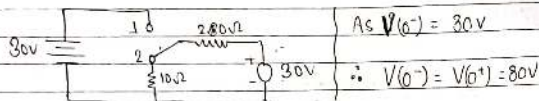
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Page: _____

42. K is moved from 1 to 2 at $t=0$, Steady State condition having reached before switching find $i(t)$, $\frac{di(t)}{dt}$ and $\frac{d^2i(t)}{dt^2}$ at $t=0^+$.



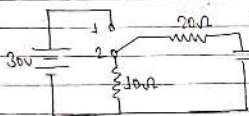
Soln

Equivalent circuit at $t=0^+$



$$i(0^+) = \frac{30}{20+10} = 1A$$

Now,



$$20i(t) + \frac{1}{C} \int i(t) dt + 10i(t) = 0$$

$$30i(t) + \frac{1}{C} \int i(t) dt = 0 \quad \text{--- (1)}$$

Date: _____
Page: _____

Diff eqn ① w.r.t 't'

$$30 \frac{di(t)}{dt} + \frac{i(t)}{C} = 0 \quad \text{--- (1)}$$

$$30 \frac{di(t)}{dt} = -\frac{i(t)}{C}$$

At $t=0^+$

$$30 \frac{di(t)}{dt} = -\frac{1}{10^{-6}}$$

$$\frac{di(t)}{dt} = -\frac{10^{-6}}{30} = -3333.33 A s^{-1}$$

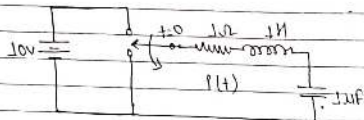
Again, diff eqn ①

$$30 \frac{d^2i(t)}{dt^2} + \frac{1}{C} \frac{di(t)}{dt} = 0$$

$$\frac{d^2i(t)}{dt^2} = -\left(\frac{-10^{-6}}{30}\right) \left(\frac{1}{10^{-6}}\right) \times \frac{1}{30}$$

$$= \frac{10^{-12}}{30^2} = \frac{10^9 \times 10^2}{-9 \times 100} = 10^9 / 9 A s^{-2}$$

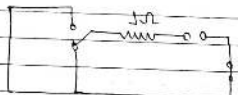
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Equivalent circuit at $t = 0^+$

A7 initial

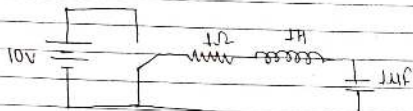


$$\int_C \vec{I} dt = 10 \text{ V}$$

$$i \in V(0') = 10V = V(0')$$

$$i(0^+) = 0A$$

Now,

$$\hat{A} \quad t=0^+$$


KVL eqⁿ is

$$i(t) + \frac{di(t)}{dt} + \frac{1}{C} \int i dt = 0 \quad \text{--- (1)}$$

$$0.1 \frac{di(t)}{dt} + 10i(t) = 0 \Rightarrow \frac{di(t)}{dt} = -10As^{-1}$$

Diff eqn ①

$$\frac{d^2 i(t)}{dt^2} + \frac{d i(t)}{dt} + i(t) = 0 \quad \text{--- (11)}$$

$$\frac{d^2 i(t)}{dt^2} = 10 \text{ A/s}^2$$

Again,

Diff eqn (11)

$$\frac{d^2 i(t)}{dt^2} + \frac{d^3 i(t)}{dt^3} + \frac{1}{C^2} \frac{di(t)}{dt} = 0$$

$$0.10 + \frac{d^3 r(t)}{dt^3} = 10 \times \frac{1}{10^{-6}} = 0$$

~~$$\frac{d^3 i(t)}{dt} = 10^7 - 10^6$$~~

$$= 9.99 \times 10^6 \text{ As}^{-3} \quad \text{IP}$$

18 In the circuit shown, the switch is moved from 1 to 2 at $t=0$. Find $i(t)$.

Using the classical method determine the solution of current differential equation

$$i''(t) + 3i'(t) + 2i(t) = 4e^t$$

$$m^2 + 3m + 2 = 0$$

$$m^2 + 2m + m + 2 = 0$$

$$\text{or } (m+1)(m+2) = 0 \quad \text{or } \frac{d}{dt} - k_1 e^{k_1 t} - 2k_2 e^{-k_2 t}$$

$$\Rightarrow m = -1, -2$$

complementary function, $i_c = k_1 e^{-t} + k_2 e^{-2t}$
for particular integral, assume trial solution as $i_p = Ae^t$

$$\frac{di_p}{dt} = Ae^t$$

Put in eqn

$$Ae^t + 3Ae^t + 2Ae^t = 4e^t$$

$$6Ae^t = 4e^t$$

$$A = \frac{4}{6}$$

$$\text{So, } i_p = \frac{4}{6} e^t$$

Total solution $i(t) = i_c + i_p$

$$= k_1 e^{-t} + k_2 e^{-2t} + \frac{4}{6} e^t \dots \textcircled{2}$$

Now, initially the given is $i(0) = 1$

$$\text{So, } 1 = k_1 e^0 + k_2 e^0 + \frac{4}{6} e^0$$

$$k_1 + k_2 = 1 - \frac{4}{6} = \frac{2}{6} \dots \textcircled{1}$$

Again, given $\frac{di}{dt}(0) = -1$

Put in eqn ②

$$-1 = -k_1 e^0 - 2k_2 e^0 + \frac{4}{6} e^0$$

$$\Rightarrow -k_1 - 2k_2 = -1 - \frac{4}{6}$$

$$\text{or, } -k_1 - 2k_2 = -1 - \frac{4}{6}$$

$$\text{or, } -k_1 - 2k_2 = -\frac{10}{6} = -\frac{5}{3} \dots \textcircled{5}$$

from ① and ⑤

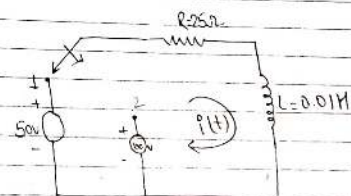
$$k_1 + k_2 = \frac{1}{3}, \quad \text{or } k_2 = \frac{1}{3} - k_1$$

$$-k_1 - 2k_2 = -\frac{5}{3}$$

$$k_2 = \frac{1}{3}$$

$$k_1 = \frac{1}{3} - k_2 = \frac{1}{3} - \frac{1}{3} = 0$$

* Switch K is in position 1 for a long time and moved to position 2 at $t=0$ find the resulting current $i(t)$ using classical method



Soln

At $t=0$, using KVL

$$Ri + L \frac{di}{dt} = 100$$

$$25i + 0.0 \frac{di}{dt} = 100$$

$$\frac{di}{dt} + \frac{25}{0.0} i = \frac{100}{0.0} = 10000$$

$$\frac{di}{dt} + 2500i = 10000$$

Comparing with $\frac{di}{dt} + Pi = Q$

where $P = 2500$, $Q = 10000$

General soln

$$= e^{-Pt} \int Q e^{Pt} dt + k e^{-Pt}$$

$$= e^{-2500t} \int 10000 e^{2500t} dt + k e^{-2500t}$$

$$= 10000 e^{-2500t} \times \frac{e^{2500t}}{2500} + k e^{-2500t}$$

$$i = 4 + k e^{-2500t}$$

Initial current

$$i(0) = i(0^+) = 50/25 = 2$$

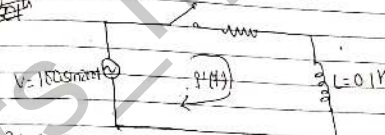
Put in eqn (1)

$$2 = 4 + k e^{-0} \Rightarrow k = -2$$

Now for particular soln

$$i(t) = 4 - 2e^{-2500t}$$

Find the total response $i(t)$ for an RL series circuit with components $R = 10 \Omega$ and $L = 0.1 H$ for sinusoidal input voltage $v(t) = 100 \sin 200t$.



At $t=0$

$$Ri + L \frac{di}{dt} = 100 \sin 200t$$

$$\text{or, } \frac{di}{dt} + \frac{10}{0.1} i = \frac{100}{0.1} \sin 200t$$

$$\text{or, } \frac{di}{dt} + 100i = 1000 \sin 200t$$

Characteristics eqn

$$\frac{di}{dt} + 100i = 0$$

$$\text{or, } m + 100 = 0$$

$$\Rightarrow m = -100$$

complementary function $i_c = k e^{-100t}$

For particular integral, the trial soln is $i_p = A \cos 200t + B \sin 200t$

$$\text{diff above eqn } \frac{di}{dt} = 200A \sin 200t + 200B \cos 200t$$

Put both in eqn ①

$$200A \sin 200t - 200B \cos 200t + 100A \cos 200t + 100B \sin 200t = 100 \sin 200t$$

$$(200A + 100B) \sin 200t + (200B - 100A) \cos 200t = 100 \sin 200t + 0 \cos 200t$$

Equating the coeff

$$200A + 100B = 100 \dots \text{①}$$

$$200B - 100A = 0 \dots \text{②}$$

Solving ① and ②

$$A = -4, B = 2$$

Particular integral, $i_p = -4 \cos 200t + 2 \sin 200t$

$$\text{Total sol}^n i(t) = i_c + i_p \\ = K e^{-100t} - 4 \cos 200t + 2 \sin 200t$$

Initially, current $i(0) = i(0^+) = 0$

Put in total solⁿ

$$0 = K e^{-100 \times 0} - 4 \cos 200 \times 0 + 2 \sin 200 \times 0 \\ = K \times 1 - 4 \times 1 + 2 \times 0$$

$$\Rightarrow K = 4$$

So, the complete solⁿ $i(t) = 4e^{-100t} - 4 \cos 200t + 2 \sin 200t$

* The total response $i(t)$ for an RC series circuit with components $R=10\Omega$ and $C=0.1\mu$ for sinusoidal input voltage $v(t)=100\sin 200t$

Laplace transform

A function $f(t)$ in time domain can be changed into a function in frequency domain

$$F(s) = \int_0^{\infty} f(t) e^{-st} dt$$

Advantages of LT

- ① With the help of Laplace Tx, the problems are greatly simplified. Simplified, simplified, simplified.
- ② With the help of Laplace Tx, the higher order integrodifferential eqⁿ can be simplified in a single step.
- ③ In respect to the initial values, the L-T eqⁿ directly incorporates the initial values in the equation so no need to specify separately.
- ④ Has many applications in communication fields and mathematical model.

Integro-differential equation	Laplace Tx	Transformed form	← Initial values
-------------------------------	------------	------------------	------------------

Time domain equation solution	Inverse L-T	← algebraic manipulation	dr rearrangement of transformation
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Time domain Frequency domain

flow diagram of L-T

Some useful L.Ts

$$\text{L.T} \left\{ \begin{matrix} u(t) \\ \text{or } 1 \end{matrix} \right\} \rightarrow \frac{1}{s}$$

$$\text{L.T} \{ t \} \rightarrow \frac{1}{s^2}$$

$$\text{L.T} \{ t^n \} \rightarrow \frac{n!}{s^{n+1}}$$

$$\text{L.T} \{ e^{at} \} \rightarrow \frac{1}{s-a}$$

$$\text{L.T} \{ e^{-at} \} \rightarrow \frac{1}{s+a}$$

$$\text{L.T} \{ \sin \alpha t \} \rightarrow \frac{\alpha}{s^2 + \alpha^2}$$

$$\text{L.T} \{ \cos \alpha t \} \rightarrow \frac{s}{s^2 + \alpha^2}$$

$$\text{L.T} \{ e^{-at} \cos \alpha t \} \rightarrow \frac{s+a}{(s+a)^2 + \alpha^2}$$

$$\text{L.T} \{ e^{-at} \sin \alpha t \} \rightarrow \frac{\alpha}{(s+a)^2 + \alpha^2}$$

Some useful properties

① Linearity

$$\text{L.T} \{ a f_1(t) + b f_2(t) \} \rightarrow a f_1(s) + b f_2(s)$$

② Differentiation

$$\text{L.T} \left\{ \frac{df(t)}{dt} \right\} \rightarrow s f(s) - f(0^+) = f'(0^+)$$

And 2nd order

$$\text{L.T} \left\{ \frac{d^2 f(t)}{dt^2} \right\} \rightarrow [s^2 f(s) - s f(0^+) - f'(0^+)]$$

Integration

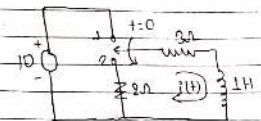
$$\text{L.T} \left[\int f(t) dt \right] = \frac{f(s)}{s} + \frac{f(0^+)}{s}$$

Ex Ri circuit

Switch K is moved from 1 to 2 at $t=0$, find the current response using L.T

Solⁿ

At $t=0$, using KVL



$$-2i(t) - 3i(t) - 1 \frac{di(t)}{dt} = 0$$

$$\frac{di(t)}{dt} + 5i(t) = 0$$

Taking L.T

$$s I(s) + 1 [5 I(s) - i(0^+)] = 0$$

$$\text{or } 5 I(s) + 5 I(s) - 10 = 0$$

$$\text{or } I_s (5+s) = 10/3 \Rightarrow I_s = \frac{10}{3(5+s)}$$

For more solved and unsolved problems visit
http://t.me/engineers_np
NCT College

Taking Inverse L.T

$$i(t) = \frac{10}{3} e^{-3t}$$

* When switch K is moved from 2 to 1

At $t=0$, using KVL;

$$1 \frac{di(t)}{dt} + 3i(t) = 10 \dots (1)$$

Taking L.T

$$s I(s) - i(0^+) + 3 I(s) = \frac{10}{s}$$

$$s I(s) - 0 + 3 I(s) = \frac{10}{s}$$

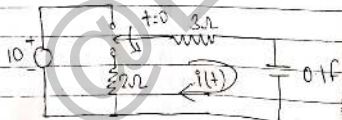
$$I_s = \frac{10}{s(s+3)}$$

$$I(s) = \frac{10}{3} \left[\frac{1}{s+3} + \frac{1}{s} \right]$$

Taking Inverse L.T

$$i(t) = \frac{10}{3} (1 - e^{-3t})$$

* Switch K is moved from 1 to 2 at $t=0$. find the current response using L.T



Q2)

At $t=0$, using KVL

$$2i(t) + 3i(t) + \frac{1}{0.1} \int i(t) dt = 0$$

$$\text{or, } 5i(t) + 10 \int i(t) dt = 0$$

Using L.T

$$5 I(s) + 10 \left[\frac{I(s)}{s} + \frac{i(0^+)}{s} \right] = 0$$

$$5 I(s) + 10 \left[\frac{I(s)}{s} + \frac{10}{3s} \right] = 0$$

$$5 I(s) + \frac{10 I(s)}{s} + \frac{100}{3s} = 0$$

$$I(s) \left[\frac{5s+10}{s} \right] = -\frac{100}{3s}$$

$$I(s) = -\frac{100}{3s} \times \frac{s}{s+10}$$

$$= -\frac{100}{3s} \times \frac{s}{s(s+2)}$$

$$= -\frac{20}{3(s+2)}$$

Taking Inverse L.T

$$i(t) = -\frac{20}{3} e^{-2t}$$

2 to 1

find the Laplace transform of unit step function, ramp function and in above question, switch k is moved from 2 to 1.

Solution,

for this case using KVL eqn

At $t=0$,

$$\frac{1}{0.1} \int i(t) dt + 3i(t) = 10$$

$$\text{or } 10 \int i(t) dt + 3i(t) = 10 \dots \text{--- (1)}$$

Taking LT on eqn (1)

$$10 \left[\frac{I(s)}{s} + \frac{i(0^+)}{s} \right] + 3I(s) = \frac{10}{s}$$

$$\text{or } 10 \left(\frac{I(s)}{s} + 0 \right) + 3I(s) = \frac{10}{s}$$

$$\text{or } \frac{10+3s}{s} I(s) = \frac{10}{s}$$

$$\text{or } I(s) = \frac{10+3s}{s} = \frac{10}{s} + 3$$

$$\Rightarrow I(s) = \frac{10}{s} + 3$$

Taking ILT

$$i(t) = \sin \left(\frac{10t}{3} \right) + 3$$

Using Laplace transformation, solve the equations

$$\frac{d^2 i(t)}{dt^2} + 4 \frac{di(t)}{dt} + 8i(t) = 8$$

$$\text{Given } i(0^+) = 3 \text{ \& } \frac{di(0^+)}{dt} = -4$$

Solution,

Given eqn is

$$\frac{d^2 i(t)}{dt^2} + 4 \frac{di(t)}{dt} + 8i(t) = 8$$

Taking LT on both sides,

$$s^2 I(s) - s i(0^+) - \frac{di(0^+)}{dt} + 4s I(s) - 4i(0^+) + 8I(s) = \frac{8}{s}$$

$$\text{or } (s^2 + 4s + 8) I(s) - 3s + 4 - 12 = \frac{8}{s}$$

$$\text{or } (s^2 + 4s + 8) I(s) = \frac{8}{s} + 3s + 8$$

$$\text{or } I(s) = \frac{3s^2 + 8s + 8}{s(s^2 + 4s + 8)}$$

Let,

$$\frac{3s^2 + 8s + 8}{s(s^2 + 4s + 8)} = \frac{A}{s} + \frac{Bs + C}{s^2 + 4s + 8}$$

$$\text{or } 3s^2 + 8s + 8 = A(s^2 + 4s + 8) + s(Bs + C)$$

When, $s=0$

$$8 = 8A \Rightarrow A = 1$$

And,

$$3s^2 + 8s + 8 = (A+B)s^2 + (4A+C)s + 8A$$

$$4A + C = 8 \Rightarrow C = 4$$

$$\text{and } A+B=3$$

$$B=2$$

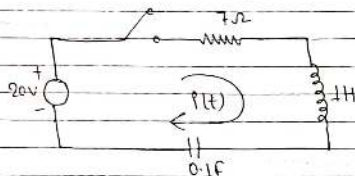
$$I(s) = \frac{1}{s} + \frac{2s+4}{s^2+4s+8}$$

Taking I.T on both sides,

$$I(t) = 1 + 2 \int_0^t \frac{8t^2}{(s+2)^2 + 2^2} dt$$

$$i(t) = 1 + 2e^{-2t} \cos 2t$$

A series of resistance 7Ω , inductor $1H$ and capacitor $0.1F$ are connected to a voltage source $20V$ as shown in figure. The elements are relaxed before switching. Find $i(t)$ using I.T



Solution

At $t=0$, using KVL

$$7i(t) + \frac{di(t)}{dt} + \frac{1}{0.1} \int i(t) dt = 20$$

$$7i(t) + \frac{di(t)}{dt} + 10 \int i(t) dt = 20 \quad \text{--- (1)}$$

$$7I(s) + [sI(s) - i(0^+)] + 10 \left[\frac{I(s)}{s} + \frac{i(0^+)}{s} \right] = \frac{20}{s}$$

$$\text{or, } I(s) (7 + s + 10/s) = 20/s$$

$$\text{or, } I(s) \left(\frac{7s + s^2 + 10}{s} \right) = \frac{20}{s}$$

$$I(s) = \frac{20}{s^2 + 7s + 10} \Rightarrow I(s) = \frac{20}{(s+2)(s+5)}$$

$$I(s) = \frac{A}{s+2} + \frac{B}{s+5}$$

$$20 = A(s+5) + B(s+2)$$

$$4 + s = -5$$

$$20 = -3B \Rightarrow B = -20/3$$

$$4 + s = -3$$

$$20 = -3A \Rightarrow A = 10$$

$$I(s) = \frac{10}{s+2} - \frac{20}{3(s+5)}$$

Taking I.T

$$i(t) = 10e^{-2t} - \frac{20}{3}e^{-5t}$$

When v is replaced by $10\sin 2t$ using KVL

$$7i(t) + \frac{di(t)}{dt} + \frac{1}{0.1} \int i(t) dt = 10\sin 2t$$

$$7i(t) + \frac{di(t)}{dt} + 10 \int i(t) dt = 10\sin 2t$$

Using I.T

$$7I(s) + sI(s) - i(0^+) + 10 \left[\frac{I(s)}{s} + \frac{i(0^+)}{s} \right] = \frac{10 \times 2}{s^2 + 4}$$

$$Q, 7I(s) + sI(s) - 0 + 10I(s) = \frac{20}{s}$$

$$Q, I(s) \left[7 + s + \frac{10}{s} \right] = \frac{20}{s^2 + 4}$$

$$I(s) = \frac{20s}{(s+2)(s+5)(s^2+4)}$$

$$\text{When } s = -5 \quad 20(-5) = -3B(29) \Rightarrow B = \frac{100}{87}$$

$$\text{When } s = -2 \quad -40 = 3A(8) \Rightarrow A = \frac{-40}{24} = -\frac{5}{3}$$

$$20s = A(s^2 + 5s^2 + 4s + 20) + B(s^2 + 2s^2 + 4s + 8) + (s+2)(s^2 + 4)$$

$$Q, 20s = A(s^2 + 5s^2 + 4s + 20) + B(s^2 + 2s^2 + 4s + 8) + C(s^3 + 7s^2 + 10s) + D(s^2 + 4s + 4)$$

$$A + B + C = 0$$

$$-\frac{5}{3} + \frac{100}{87} + C = 0 \Rightarrow C = \frac{5}{3} - \frac{100}{87} = \frac{15}{29}$$

$$20A + 8B + 10D = 0$$

$$Q, -\frac{100}{3} + \frac{800}{87} + 10D = 0$$

$$Q, 10D = \frac{100}{3} - \frac{800}{87}$$

$$\Rightarrow D = \frac{70}{29}$$

$$I(s) = \frac{-5}{3(s+2)} + \frac{100}{87(s+5)} + \frac{15s}{29} + \frac{70}{s^2+4}$$

$$= \frac{-5}{3(s+2)} + \frac{100}{87(s+5)} + \frac{15s}{29(s^2+4)} + \frac{2 \cdot 35}{s^2+4}$$

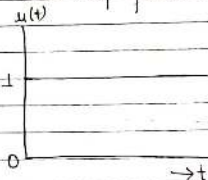
Taking Inverse LT on both sides, we get

$$i(t) = \frac{-5}{3} e^{-2t} + \frac{100}{87} e^{-5t} + \frac{15 \cos 2t + 35 \sin 2t}{29}$$

Find the Laplace transform of unit step function, ramp function and Dirac delta function

Solution,

Unit Step function



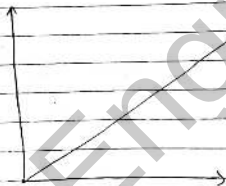
$$u(t) = \begin{cases} 1 & \text{for } t \geq 0 \\ 0 & \text{else} \end{cases}$$

$$\text{L.T. } \{u(t)\} = \int_0^{\infty} u(t) e^{-st} dt = \frac{1}{s}$$

Ramp function

$$r(t) = \begin{cases} t & \text{for } t \geq 0 \\ 0 & \text{else} \end{cases}$$

$$\text{L.T. } \{r(t)\} = \int_0^{\infty} t e^{-st} dt = \frac{1}{s^2}$$



Dirac Delta function

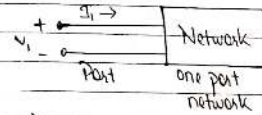
or Unit Impulse function



$$\delta(t) = \begin{cases} 1 & \text{at } t=0 \\ 0 & \text{else} \end{cases}$$

$$\text{L.T. } \{\delta(t)\} = 1$$

One port and two port network



Driving point impedance $Z_{in} = \frac{v_i(s)}{i_i(s)}$

Transformed impedance and admittance

① for resistor

$$V(t) = i(t) R$$

Taking LT

$$V(s) = I(s) R$$

$$Z(s) = R = \frac{V(s)}{I(s)} = \text{Impedance}$$

$$\text{Admittance } Y(s) = \frac{I(s)}{V(s)}$$

② Inductor

$$V(t) = L \frac{di}{dt}$$

Taking LT

$$V(s) = L [sI(s) - i(0^+)]$$

$$= LsI(s) - Li(0^+)$$

$$\text{let, } V_1(s) = LsI(s)$$

$$\frac{V_1(s)}{I(s)} = Ls = \text{Impedance } Z(s)$$

$$Y(s) \text{ Admittance} = \frac{1}{Ls}$$

③ for capacitor

$$V(t) = \frac{1}{C} \int i(t) dt$$

Taking LT

$$V(s) = \frac{1}{C} \left[\frac{I(s)}{s} + \frac{i(0^+)}{s} \right]$$

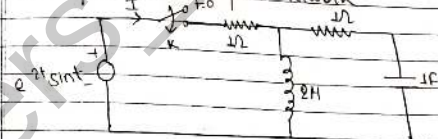
$$V(s) = \frac{I(s)}{Cs} + \frac{i(0^+)}{Cs}$$

$$\text{let, } V_1 = \frac{I(s)}{Cs}$$

$$\frac{V_1}{I(s)} = \frac{1}{Cs} = \text{Impedance } Z(s)$$

$$\text{Admittance } Y(s) = Cs$$

One port and two port network



Switch K is closed at $t=0$. find I .

for

Applying transformed form

$$\text{In series } 1 + \frac{1}{s} = \frac{s+1}{s}$$

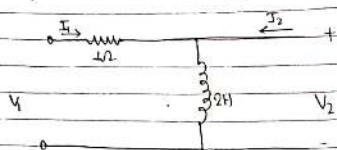
$$\text{Again, } \left(\frac{s+1}{s} \right) \parallel 2s$$

$$= \frac{\frac{s+1}{s} \times 2s}{\frac{s+1}{s} + 2s}$$

$$= \frac{s+1}{s+2s+1}$$

$$= \frac{2(s+1)s}{2s^2+2s+1}$$

Example: Find $G_{21}(s)$, $Z_{21}(s)$ and $Z_{11}(s)$



Solution. Two port network

Using KVL

$$V_1(s) = I_1(s) \times 1 + 2s \times I_1(s)$$

$$V_1(s) = I_1(s) [1 + 2s]$$

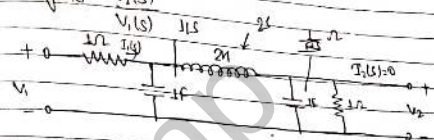
$$V_2(s) = I_1(s) \times 2s$$

$$G_{21}(s) = \frac{V_2(s)}{V_1(s)} = \frac{I_1(s) \times 2s}{I_1(s) [1 + 2s]} = \frac{2s}{1 + 2s}$$

$$Z_{21}(s) = \frac{V_2(s)}{I_1(s)} = 2s \Omega$$

$$Z_{11} = \frac{V_1(s)}{I_1(s)} = (1 + 2s) \Omega$$

Example Find $V_1(s)$



Solution,

$$V_1(s) = I_1(s) \times 1 + \frac{1}{1+s} [I_1(s) - I_3(s)]$$

$$V_1(s) = I_1(s) \left[1 + \frac{1}{1+s} \right] - \frac{1}{1+s} I_3(s) \dots \text{--- (1)}$$

Using KVL in 2nd loop,

$$I_3(s) \times 2s + \frac{1}{1+s} I_3(s) + \frac{1}{s} [I_1(s) - I_3(s)] = 0$$

$$\text{or, } I_3(s) \left[2s + \frac{1}{1+s} + \frac{1}{s} \right] - \frac{1}{s} I_1(s) = 0 \dots \text{--- (2)}$$

Again,

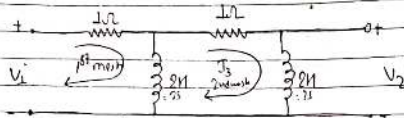
$$V_2(s) = \frac{1}{1+s} \times I_3(s) \dots \text{--- (3)}$$

$$I_3(s) = V_2(s) \times (1+s)$$

Put in eqn (1)

$$V_1(s) = I_1(s) \left[1 + \frac{1}{1+s} \right] - \frac{1}{s} \times V_2(s) (1+s)$$

Find the voltage transfer function $V_2(s)$ in the given circuit. Assume the circuit is terminated $V_2(s)$ by 2Ω . Then find $X_1(s)$, $Z_1(s)$, $Z_2(s)$.



Solution.

From 1st mesh, using KVL

$$V_1(s) = I_1(s) \times 1 + 2s [I_1(s) - I_3(s)]$$

$$V_1(s) = (1+2s)I_1(s) - 2sI_3(s) \dots \text{--- (1)}$$

From 2nd mesh, using KVL

$$1 \times I_3(s) + 2sI_3(s) + 2s[I_3(s) - I_1(s)] = 0$$

$$I_3(s)[1+4s] - 2sI_1(s) = 0 \dots \text{--- (2)}$$

And,

$$V_2(s) = 2s \times I_3(s) \dots \text{--- (3)}$$

From equation (3), $I_3(s) = \frac{V_2(s)}{2s}$

From equation (1),

$$V_1(s) = (1+2s)I_1(s) - 2s \times \frac{V_2(s)}{2s}$$

$$V_1(s) = (1+2s)I_1(s) - V_2(s) \dots \text{--- (4)}$$

From (2) $I_3(s) = \frac{I_1(s)(1+4s)}{2s}$

Put in eqn (4)

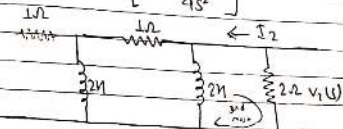
$$V_1(s) = (1+2s) \frac{(1+4s)}{2s} I_3(s) - V_2(s)$$

$$= \frac{(1+2s)(1+4s)}{2s} \times \frac{V_2(s)}{2s} - V_2(s)$$

$$V_1(s) = \frac{V_2(s)}{4s^2} [1+6s+8s^2+4s^2]$$

$$= V_2(s) = \frac{4s^2+6s+1}{4s^2}$$

2nd case



In 1st mesh, using KVL

$$V_1(s) = I_1(s) \times 1 + 2s [I_1(s) - I_3(s)]$$

$$V_1(s) = (1+2s)I_1(s) - 2sI_3(s) \dots \text{--- (1)}$$

In 2nd mesh, using KVL

$$1 \times I_3 + 2s[I_3(s) + I_2(s)] + 2s[I_3(s) - I_1(s)] = 0$$

$$\text{Or, } (1+4s)I_3(s) + 2sI_2(s) - 2sI_1(s) = 0 \dots \text{--- (2)}$$

In 3rd mesh, using KVL

$$2s[I_2(s) + I_3(s)] + 2I_2(s) = 0$$

$$\text{Or, } (2+2s)I_2(s) + 2sI_3(s) = 0 \dots \text{--- (3)}$$

From eqn (3)

$$I_3(s) = -\frac{(2+2s)I_2(s)}{2s} = -\left(\frac{1+s}{s}\right)I_2(s)$$

Equation ② becomes

$$-(1+4s) \frac{(1+s)}{s} I_1(s) + 2s [I_1(s)] - 2s I_1(s) = 0$$

$$\text{Or, } I_2(s) \left[\frac{-(1+4s)(1+s)}{s} + 2s \right] - 2s I_1(s)$$

$$\text{Or, } I_2(s) \left[\frac{-(1+4s+s+4s^2+2s^2)}{s} \right] - 2s I_1(s)$$

$$\frac{I_2(s)}{I_1(s)} = \frac{-2s^2}{6s^2+5s+1} \quad \#$$

Also

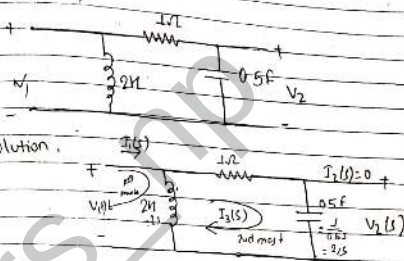
$$V_2(s) = -2I_2(s) \dots \text{⑤}$$

$$\text{From ⑤ } I_2(s) = \frac{-V_2(s)}{2}$$

Eqⁿ ② becomes,

$$(1+4s) I_2(s) - s V_2(s) - 2s I_1(s) = 0 \dots \text{⑥}$$

Find $\frac{V_2(s)}{V_1(s)}$



Solution.

In 1st mesh, using KVL

$$V_1(s) - 2s [I_1(s) - I_2(s)] - V_2(s) = 0$$

In 2nd mesh, using KVL

$$1 \times I_2(s) + \frac{2}{s} I_2(s) + 2s [I_2(s) - I_1(s)] = 0$$

$$I_2(s) \left[1 + \frac{2}{s} + 2s \right] - 2s I_1(s) = 0 \dots \text{②}$$

$$\text{And, } V_2(s) = \frac{2}{s} I_2(s) \dots \text{③}$$

From eqⁿ ③

$$I_2(s) = \frac{s}{2} V_2(s)$$

$$\text{From ② } I_1(s) = \frac{1 + \frac{2}{s} + 2s}{2s} I_2(s)$$

Put in equation ①, we get $\frac{V_1(s)}{V_1(s)}$

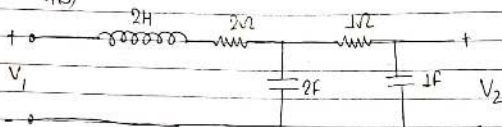
$$V_1(s) = 2s \left(\frac{s+2+2s^2}{2s^2} \right) I_3(s) - 2s I_3(s)$$

$$= \frac{[2s^2+s+2]}{s} I_3(s) - 2s I_3(s) = \frac{s+2}{s} I_3(s)$$

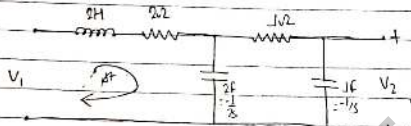
$$\Rightarrow \frac{V_2(s)}{V_1(s)} = \frac{2I_3}{s(s+2)I_3(s)}$$

$$\frac{V_2(s)}{V_1(s)} = \frac{2}{s+2}$$

* Find $\frac{V_2(s)}{V_1(s)}$



Solution.



Using KVL in first mesh

$$V_1(s) = 2s I_1(s) + 2 I_1(s) + \frac{1}{2s} [I_1(s) - I_3(s)]$$

$$V_1(s) = I_1(s) \left(2s + 2 + \frac{1}{2s} \right) - I_3(s) \frac{1}{2s}$$

Using KVL in 2nd mesh

$$1 \times I_3(s) + \frac{1}{2s} I_3(s) + \frac{1}{2s} [I_3(s) - I_1(s)] = 0$$

$$Or, I_3(s) \left[1 + \frac{1}{2s} + \frac{1}{2s} \right] - I_1(s) \frac{1}{2s} = 0$$

$$Or, I_3(s) \left(1 + \frac{1}{s} \right) - I_1(s) \frac{1}{2s} = 0 \dots \dots \textcircled{2}$$

Using KVL in 3rd mesh

$$V_2(s) = \frac{I_3(s)}{s} \dots \dots \textcircled{3}$$

From equation ②,

$$I_1(s) = \left(\frac{s+1}{s} \right) \times 2s = 2(s+1) I_3(s)$$

$$\Rightarrow I_1(s) = 2(s+1) I_3(s) \dots \dots \textcircled{4}$$

From eqⁿ ①

$$\frac{V_1(s)}{V_1(s)} = I_3(s) \cdot 2 \left(2s + 2 + \frac{1}{2s} \right) (s+1) - I_3(s) \frac{1}{2s}$$

$$= I_3(s) \left[\frac{(4s^2 + 2s + 1)(s+1)}{s} \right] - I_3(s) \frac{1}{2s}$$

$$V_1(s) = I_3(s) \left[\frac{2(s+1)(4s^2 + 2s + 1) - 1}{2s} \right] \dots \dots \textcircled{5}$$

Dividing eqn (4) and (5)

$$\frac{V_2(s)}{V_1(s)} = \frac{T_3(s)}{s \cdot 2s \cdot T_3(s)} [2(s+1)(4s^2+2s+1)-1]$$

$$\begin{aligned} \frac{V_2(s)}{V_1(s)} &= \frac{2(s+1)(4s^2+2s+1)-1}{2s^2} \\ &= \frac{8s^3+4s^2+2s+8s^2+4s+2-1}{2s^2} \\ &= \frac{8s^3+12s^2+6s+1}{2s^2} \quad \text{IP} \end{aligned}$$

Network function Representation

say $F(s) = \frac{P(s)}{Q(s)} = \frac{a_0s^n + a_1s^{n-1} + a_2s^{n-2} + \dots + a_{n-1}s + a_n}{b_0s^m + b_1s^{m-1} + b_2s^{m-2} + \dots + b_{m-1}s + b_m}$

where $a_0, a_1, a_2, \dots, a_n$ and $b_0, b_1, b_2, \dots, b_m$ are the positive coefficients. The numerator polynomial has n roots $z_1, z_2, z_3, \dots, z_n$ and denominator has roots $p_1, p_2, p_3, \dots, p_m$. So, the polynomial can be written as:

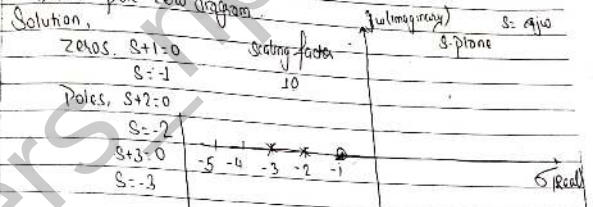
$$F(s) = \frac{P(s)}{Q(s)} = \frac{(s-z_1)(s-z_2)(s-z_3)\dots(s-z_n)}{(s-p_1)(s-p_2)(s-p_3)\dots(s-p_m)}$$

If any root in the numerator equals s , the whole polynomial will be 0. So, the roots $z_1, z_2, \dots, z_n$ are known as zero. Similarly, if any roots in the denominator equals s , the

whole polynomial will be infinity. So the roots $p_1, p_2, \dots, p_m$ are known as poles.

$$T(s) = \frac{10(s+1)}{(s+2)(s+3)}$$

Show pole-zero diagram



Restrictions in the location of poles and zeros

- 1) All the roots must be positive and real
- 2) These roots may be complex but must occur in conjugate
- 3) There should be the continuous orders of s in the numerator and denominator i.e. should not have missing component
- 4) The highest power in numerator and denominator must not differ more than by unity
- 5) The lowest power in numerator and denominator must not differ more than by unity
- 6) All the poles and zeros lie on the left half of s -plane

Draw the pole zero diagram of the network function

$$T(s) = \frac{3s}{(s+1)(s+2)}$$

Find the time domain response from pole zero diagram

Solution,

$$\text{zero, } s=0 \text{ (origin)}$$

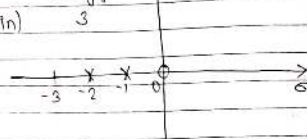
$$\text{Poles, } s+1=0$$

$$\Rightarrow s=-1$$

$$s+2=0$$

$$\Rightarrow s=-2$$

scaling factor



$$T(s) = \frac{3s}{(s+1)(s+2)}$$

$$\frac{3s}{(s+1)(s+2)} = \frac{k_1}{s+1} + \frac{k_2}{s+2}$$

$$K_1 = \frac{3(-1-0)}{[-1-(-2)]} \quad [\text{if } -1 \text{ is origin } k_1 = 3(-1-(-2)) = 3(-1+2)]$$

$$K_2 = \frac{-3}{[-2-(-1)]} = 6$$

Then,

$$T(s) = \frac{-3}{s+1} + \frac{6}{s+2}$$

Taking Laplace

$$i(t) = -3e^{-t} + 6e^{-2t} \quad \text{TR}$$

$$T(s) = \frac{3s}{(s+2)(s^2+2s+2)}$$

$$\text{zero, } s=0 \text{ (origin)}$$

$$\text{poles, } s+2=0$$

$$\Rightarrow s=-2$$

$$s^2+2s+2=0$$

$$\text{or, } s^2+2s+1-1+2=0$$

$$\text{or, } (s+1)^2 - 1 + 2 = 0$$

$$\text{or, } (s+1)^2 = -1 = -j^2$$

$$\text{or, } (s+1) = \pm j$$

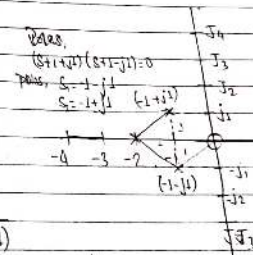
$$\text{or, } s = -1 \pm j$$

$$s_{1,2} = -2 \pm \sqrt{4-4} \cdot 1 \cdot 2$$

$$= -2 \pm \sqrt{-4}$$

$$= -2 \pm j2$$

$$= -1 \pm j1$$



$$T(s) = \frac{3s}{(s+2)(s^2+2s+2)}$$

$$= \frac{k_1}{s+2} + \frac{k_2}{s+1+j} + \frac{k_3}{s+1-j}$$

numerator $\rightarrow$ zero
denominator $\rightarrow$

$$K_1 = \frac{3(-2-0)}{[-2-(-1-j)]} = \frac{-6}{-1-j} = -3$$

$$K_2 = \frac{3 \times \sqrt{2}}{\sqrt{2} \times 2} = \frac{3}{2}$$

$$K_3 = \frac{3}{2}$$

$$T(s) = \frac{-3}{s+2} + \frac{3/2}{s+1+j1} + \frac{3/2}{s+1-j1}$$

$$= \frac{-3}{s+2} + \frac{3}{2} \left[\frac{s+1-j1+s+1+j1}{(s+1+j1)(s+1-j1)} \right]$$

$$= \frac{-3}{s+2} + \frac{3}{2} \frac{2s+2}{(s+1)^2+1^2}$$

$$= \frac{-3}{s+2} + \frac{3}{2} \times \frac{2(s+1)}{(s+1)^2+1^2}$$

Taking ILT

$$i(t) = -3e^{-2t} + 3e^{-t} \cos t \quad \text{IP}$$

Stability of network

To check the stability of a network, we use Routh's stability criterion. According to it, all the poles and zeros must lie on the left half of s-plane.

Let, a network described by a polynomial

$$P(s) = a_0 s^n + a_1 s^{n-1} + a_2 s^{n-2} + a_3 s^{n-3} + \dots + a_{n-1} s + a_n$$

We can apply Routh's stability array criterion to check whether the network described by above polynomial is stable or not.

Routh's table

s^n	a_0	a_2	a_4	a_6
s^{n-1}	a_1	a_3	a_5	a_7
s^{n-2}	b_1	b_2	b_3	
s^{n-3}	c_1	c_2	c_3	
$\vdots$				
s^0				

$$b_1 = \frac{a_1 a_2 - a_0 a_3}{a_1}, \quad b_2 = \frac{a_1 a_4 - a_0 a_5}{a_1}, \quad b_3 = \frac{a_1 a_6 - a_0 a_7}{a_1}$$

$$c_1 = \frac{b_1 a_3 - a_1 b_2}{b_1}, \quad c_2 = \frac{b_1 a_5 - a_1 b_3}{b_1}$$

Conclusion: If all the elements in the 1st column is positive, the network is stable otherwise the network is unstable.

The number of sign changes indicates the number of roots lie on the right half of s-plane

A network is described by a polynomial

$$P(s) = s^3 + 2s^2 + 2s + 40$$

Check for stability using Routh's array

Solution.

$$P(s) = s^3 + 2s^2 + 2s + 40$$

Routh's table.

s^3	1	2	0
s^2	2	40	
s^1	$\frac{4-40}{2} = -18$	0	
s^0	40		

Since the coefficient of the first column of Routh table is negative, the given network is unstable

$$P(s) = s^4 + s^3 + 2s^2 + 2s + 3$$

Check for stability.

Soln

$$P(s) = s^4 + s^3 + 2s^2 + 2s + 3$$

Routh table

s^4	1	2	3
s^3	1	2	0
s^2	$\frac{2-2}{1} = 0$	$\frac{3-2}{1} = 1$	0
s^1	∞		
s^0			

Case When 1st element comes as zero
let α be the smallest positive element

s^4	1	2	3
s^3	1	2	0
s^2	$\frac{2\alpha-2}{\alpha}$	$\frac{3-2\alpha}{\alpha}$	0
s^1	$\frac{(2\alpha-3)/\alpha}{\alpha}$	0	
s^0	3		

The system is stable if

$$\frac{2\alpha-3}{\alpha} > 0$$

$$\text{or } 2\alpha - 3 > 0$$

$$\Rightarrow \alpha > \frac{3}{2} = 1.5$$

The system is stable if $\alpha > 1.5$ and unstable $\alpha \leq 1.5$

$$P(s) = s^3 + 2s^2 + 2s + A$$

find the range of A so that the network is stable

Soln

$$P(s) = s^3 + 2s^2 + 2s + A$$

Routh table

s^3	1	2	0
s^2	2	A	0
s^1	$\frac{4-A}{2}$	0	
s^0	$\frac{4-A}{2}$		

$$\frac{4-A}{2} > 0$$

$$\text{or, } 4-A > 0$$

$$\Rightarrow 4 > A$$

Hence, the system is stable if A is less than 4 i.e. $A < 4$

$$P(s) = s^5 + s^4 + s^3 + s^2 + s + 1$$

Solution

s^5	1	1	1
s^4	1	1	1
s^3	0	0	0
s^2			
s^1			
s^0			

Since, the elements in third row are zero, so we've to move another polynomial from second row.

$$Q(s) = 1s^4 + 1s^3 + 1s^2$$

$$R(s) = 4s^3 + 3s^2 + 2s$$

Routh table

s^5	1	1	1
s^4	1	1	1
s^3	4	3	2
s^2	$\frac{4 \cdot 3 - 1 \cdot 1}{4} = \frac{11}{4}$	$\frac{4 \cdot 2 - 1 \cdot 1}{4} = \frac{7}{4}$	
s^1	$\frac{\frac{11}{4} \cdot \frac{7}{4} - 4 \cdot 2}{\frac{11}{4}} = -5$	2	
s^0	2		

Since the coefficient of first column is negative so the network is unstable.

Fourier Series

Any periodic function can be represented by sine and cosine functions by using Fourier Series. This series is given by

$$f(t) = a_0 + \sum_{n=1}^{\infty} (a_n \cos n\omega t + b_n \sin n\omega t)$$

$$\text{where, } a_0 = \frac{1}{T} \int_0^T f(t) dt$$

$$a_n = \frac{2}{T} \int_0^T f(t) \cos n\omega t dt$$

$$b_n = \frac{2}{T} \int_0^T f(t) \sin n\omega t dt$$

where, n is any integer

$$\omega_0 = \text{fundamental frequency} = \frac{2\pi f_0}{T}$$

$T \leftarrow$ Time period of waveform

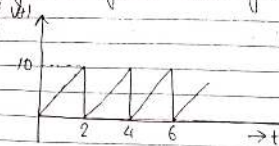
Some useful integrations

For any integers m and n , we observe that

$$\int_0^T \sin m\omega t dt = 0 \text{ for all } m$$

$$\int_0^T \cos n\omega t dt = 0 \text{ for all } n$$

Find the Fourier Series of waveform



Solution

According to f.s

$$f(t) = a_0 + \sum_{n=1}^{\infty} (a_n \cos n\omega t + b_n \sin n\omega t)$$

where, $a_0 = \frac{1}{T} \int_0^T f(t) dt$

Time period $T = 4$

$y = v(t)$
 $x = t$

$$f(t) = \frac{y_2 - y_1}{x_2 - x_1} (x - x_1)$$

$$\text{Or } v(t) - 0 = \frac{10 - 0}{2 - 0} (t - 0)$$

$$\text{Or } v(t) = 5t \quad 0 \leq t \leq 2$$

Then, $a_0 = \frac{1}{T} \int_0^T 5t dt$

$$= \frac{5}{2} \left[\frac{t^2}{2} \right]_0^2$$

$$= \frac{5}{2} \times 4 = 5$$

$$\text{And, } a_n = \frac{2}{T} \int_0^T f(t) \cos n\omega t dt$$

$$= \frac{2}{2} \int_0^2 5t \cos n \frac{2\pi}{4} t dt$$

$$= 5 \int_0^2 t \cos n \frac{2\pi}{4} t dt$$

$$= 5 \int_0^2 t \cos n\pi t dt$$

$$= \frac{5}{n\pi} \left[\sin n\pi t \right]_0^2 - \int_0^2 \frac{dt}{dt} \left[\frac{t^2}{2} \cos n\pi t \right] dt$$

$$= 5 \left[\frac{\sin n\pi t}{n\pi} \right]_0^2 - \int_0^2 \frac{\sin n\pi t}{n\pi} dt$$

$$= 5 \left[\frac{1}{n\pi} (2 \sin 2n\pi - 0) + \frac{1}{n^2\pi^2} [\cos n\pi t]_0^2 \right]$$

$$= 5 \left[\frac{2}{n\pi} (\sin 2n\pi) + \frac{1}{n^2\pi^2} (\cos 2n\pi - 1) \right]$$

$$= 5 \left[0 + \frac{1}{n^2\pi^2} (\cos 2n\pi - 1) \right]$$

$$= 5 \left(\frac{\cos 2n\pi}{n^2\pi^2} - \frac{1}{n^2\pi^2} \right)$$

$$= 0$$

Also,

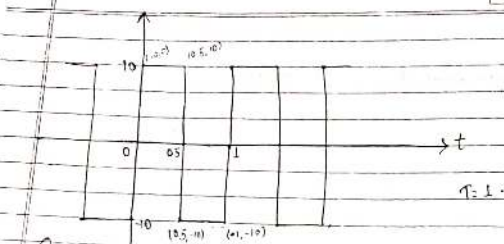
$$b_n = \frac{2}{T} \int_0^T f(t) \sin n\omega t dt$$

$$\begin{aligned}
 &= \int_0^{\pi} 5t \sin(n\pi t) dt \\
 &= 5 \int_0^{\pi} t \sin(n\pi t) dt \\
 &= 5 \left[t \int_0^{\pi} \sin(n\pi t) dt - \int_0^{\pi} t \frac{d}{dt} \int_0^{\pi} \sin(n\pi t) dt \right] \\
 &= 5 \left[-t \left(\frac{\cos(n\pi t)}{n\pi} \right) + \int_0^{\pi} \left(\frac{\cos(n\pi t)}{n\pi} \right) dt \right] \\
 &= 5 \left[-\frac{1}{n\pi} [2\cos 2n\pi - 0] + \left[\frac{\sin(n\pi t)}{n^2\pi^2} \right]_0^{\pi} \right] \\
 &= 5 \left[-\frac{1}{n\pi} \times 2 \right] = -\frac{10}{n\pi}
 \end{aligned}$$

$$\begin{aligned}
 \therefore f(t) &= 5 + \sum_{n=1}^{\infty} \left[0 + \frac{-10}{n\pi} \sin n\omega t \right] \\
 &= 5 - \frac{10}{\pi} \sum_{n=1}^{\infty} \left(\frac{1}{n} \sin n\omega t \right) \quad \text{---}
 \end{aligned}$$

Some useful integration

- ③ $\int_0^T \sin m\omega t \cos n\omega t dt = 0$ for all $m \neq n$.
- ④ $\int_0^T \sin m\omega t \sin n\omega t dt = 0$ for $m \neq n$.
- ⑤ $\int_0^T \cos m\omega t \cos n\omega t dt = 0$ for $m \neq n$.
- ⑥ $\int_0^T \sin^2 m\omega t dt = \frac{T}{2}$ for m .
- ⑦ $\int_0^T \cos^2 n\omega t dt = \frac{T}{2}$ for n .



Solution.

$$f(t) = \begin{cases} 10 & 0 \leq t \leq 0.5 \\ -10 & 0.5 \leq t \leq 1 \end{cases}$$

$$F(t) = a_0 + \sum_{n=1}^{\infty} (a_n \cos n\omega t + b_n \sin n\omega t)$$

$$a_0 = \frac{1}{T} \int_0^T f(t) dt$$

$$= \frac{1}{1} \left[\int_0^{0.5} 10 dt - \int_{0.5}^1 10 dt \right]$$

$$= \frac{1}{1} [10(0.5-0) - 10(1-0.5)]$$

$$= 0$$

$$a_n = \frac{2}{T} \int_0^T f(t) \cos n\omega t dt$$

$$= \frac{2}{1} \left[\int_0^{0.5} 10 \cos n\omega t dt - \int_{0.5}^1 10 \cos n\omega t dt \right]$$

$$= 2 \left[10 \int_0^{0.5} \cos \left(\frac{2\pi}{T} \cdot n \cdot t \right) dt - 10 \int_{0.5}^1 \cos \left(\frac{2\pi}{T} \cdot n \cdot t \right) dt \right]$$

$$= 2 \left[0 - 10 \int_{0.5}^1 \cos(2n\pi t) dt \right]$$

$$= -20 \left[\frac{\sin 2n\pi t}{2n\pi} \right]_{0.5}^1$$

$$= -\frac{20}{2n\pi} [\sin 2n\pi - \sin n\pi]$$

Even signal

$$f(t) = f(-t)$$

Eg $\cos t$



$$a_0 = \frac{1}{T} \int_0^T f(t) dt$$

$$a_n = \frac{2}{T} \int_0^T f(t) \cos n\omega t dt$$

$$b_n = 0$$

Odd signal

$$f(t) = -f(-t)$$

Eg $\sin t$

$$a_0 = 0$$

$$a_n = 0$$

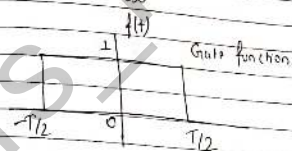
$$b_n = \frac{2}{T} \int_0^T f(t) \sin n\omega t dt$$

Fourier Transform

for non-periodic waveform
if a function $f(t)$ is in time domain, its Fourier transform is given by

$$F(\omega) = \int_{-\infty}^{\infty} f(t) e^{-j\omega t} dt$$

Eg.



Gate function

$$\frac{e^{j\omega T/2} - e^{-j\omega T/2}}{2j} = \sin$$

$$\frac{e^{j\omega T/2} + e^{-j\omega T/2}}{2} = \cos$$

Solution.

$$f(t) = 1 \quad -T/2 \leq t \leq T/2$$

$$F(\omega) = \int_{-T/2}^{T/2} 1 \cdot e^{-j\omega t} dt$$

$$= \left[\frac{e^{-j\omega t}}{-j\omega} \right]_{-T/2}^{T/2}$$

$$= \frac{1}{j\omega} \left[e^{-j\omega T/2} + e^{j\omega T/2} \right]$$

$$= \frac{1}{j\omega} \left[e^{j\omega T/2} - e^{-j\omega T/2} \right]$$



$$f(t) = 1 \quad -1 \leq t \leq 0$$

$$-1 \quad 0 \leq t \leq 1$$

$$F(\omega) = \int_{-1}^0 1 e^{j\omega t} dt + \int_0^1 -1 e^{j\omega t} dt$$

Positive real function (PRF)

A network function must be positive real function so that it can be physically realised.

To be PRF, the function $F(s)$ must possess the given properties.

- ① If $F(s)$ is PRF, then $1/F(s)$ is also PRF.
 - ② All the poles and zeros of the function $F(s)$ should be on the left half of s-plane.
 - ③ There should be no repeated poles and zeros.
 - ④ The sum of PR functions is also PR.
 - ⑤ Only simple poles can be located on $j\omega$ axis whose residues should be positive.
 - ⑥ The highest powers of numerator and denominator should be differentiated by unity.
 - ⑦ The same happens with lowest powers.
- The necessary and sufficient conditions for $F(s)$ to be PRF:
- 1) $F(s)$ must not have poles on right half of s-plane.
 - 2) $F(s)$ may have only simple poles on $j\omega$ axis with real and positive residues.
 - 3) $\operatorname{Re} [F(j\omega)] \geq 0$

even
odd

Date
Page

To test whether a given function is PRF or not.

① If $F(s)$, $P(s)$, then $P(s)$ and $Q(s)$ both sides should be Hurwitz $Q(s)$

a) This can be checked either by continued division of the even terms by odd terms of the polynomial or

b) Checking the 1st column of Routh's array. They should be positive

② If the poles lie on the jw axis, the residues should be positive

③ Apply $m_1, m_2, n_1, n_2 \mid \geq 0$ should be satisfy.

Check whether the given function is PRF or not

$$F(s) = \frac{(s+1)(s+3)}{(s+2)(s+4)}$$

Solution.

$$P(s) = (s+1)(s+3) = s^2 + 4s + 3$$

Routh's table is

s^2	1	3
s^1	4	0
s^0	3	

$P(s)$ is Hurwitz (all positive)

$$Q(s) = (s+2)(s+4) = s^2 + 6s + 8$$

Routh's table is

s^2	1	8
s^1	6	0
s^0	8	

$Q(s)$ is Hurwitz (all positive)

④ No need to check for residues.

$$\text{⑤ } m_1 = s^2 + 3, \quad m_2 = s^2 + 8$$

$$n_1 = 4s, \quad n_2 = 6s$$

Then,

$$m_1, m_2, n_1, n_2 \mid s=jw$$

$$a_1 = (s^2 + 3)(s^2 + 8) - 4s \times 6s \mid s=jw$$

$$a_1 = s^4 + 11s^2 + 24 - 24s^2 \mid s=jw$$

$$a_1 = (jw)^4 + 11(jw)^2 + 24 - 24(jw)^2 \mid s=jw$$

$$a_1 = w^4 - 11w^2 + 24 + 24w^2$$

$$= w^4 + 13w^2 + 24 \geq 0$$

∴ This follows all the properties so it is PRF.

Check for PRF
 (i) $Z(s) = \frac{2s^2+2s+4}{s^3+2s^2+s+2}$

Solution
 $P(s) = 2s^2+2s+4$

Routh's table is

s^2	2	4
s	2	0
s^0	2	

$P(s)$ is Hurwitz

$Q(s) = s^3+2s^2+s+2$

Routh's table is

s^3	1	2	0	}
s^2	2	2	0	
s				
s^0				

Even function = $2s^2+2$

odd function = s^3+s

$(s^3+s) \div (2s^2+2) = \frac{1}{2}s + \frac{1}{2}$

0 premature condition.

$Q(s)$ is Hurwitz

$\frac{s^3+s}{2s^2+2} = \frac{s}{2} \left(\frac{s^2+1}{s^2+1} \right) = \frac{s}{2}$

$f(s) = \frac{s^3+2s^2+s+2}{(2s^2+2) \left(\frac{s}{2} + 1 \right)}$

Since, $\left(\frac{s}{2} + 1 \right)$ is Hurwitz

So, s^3+2s^2+s+2 is also Hurwitz

(ii) Here,

Poles lie in jw axis because

$s^3+2s^2+s+2=0$

$0 \div s^2(s+2) + 1(s+2) = 0$

$0 \div (s+2)(s^2+1) = 0$

$\Rightarrow s = \pm j, -2$

$\frac{2s^2+2s+4}{s^3+2s^2+s+2} = \frac{k_1}{s+2} + \frac{k_2s+k_3}{s^2+1}$

$0 \div 2s^2+2s+4 = k_1(s^2+1) + (k_2s+k_3)(s+2)$

$\Rightarrow k_1 = 5/8, k_2 = 11/8, k_3 = 27/16$

$0 \div k_1+k_2+k_3 \geq 0$

(iii) $m_1 = 2s^2+4, m_2 = 2s^2+2$

$n_1 = 2s, n_2 = s^3+s$

Now, $m_1m_2 - n_1n_2 \mid s=jw$

$= (2s^2+4)(2s^2+2) - 2s(s^3+s) \mid s=jw$

Q. (2s²+4)
= 4(s²+2)(s²+1) - 2s(s²+5) | s=jw

= 4(s⁴+s²+2s²+2) - 2s³-2s² | s=jw

= 4s⁴+12s²+8-2s³-2s² | s=jw

= 4s⁴+10s²-2s³+8 | s=jw

= 4(jw)⁴+10(jw)²-2(jw)³+8

= 4w⁴-10w²-2jw³+8

= 4w⁴-10w²+2w³(j)+8 > 0

* Z(s) = $\frac{s^2+6s+8}{s^2+4s+3}$

Solution.

Given function is

Z(s) = $\frac{s^2+6s+8}{s^2+4s+3} = \frac{(s+4)(s+2)}{(s+3)(s+1)}$

① Checking for Hurwitz

P(s) = s²+6s+8

Now, Routh's table is

s ²	1	6	0
s ¹	6	0	
s ⁰	8		

P(s) is Hurwitz

Q(s) = s²+4s+3

Routh's table is,

s ²	1	3	0
s ¹	4	0	
s ⁰	3		

Q(s) is Hurwitz

The given function is Hurwitz.

(ii) Here poles are not lie on the imaginary axis so second test is not required

(iii) Here,

m₁ = s²+8

n₁ = 6s

m₂ = s²+3

n₂ = 4s

m₁m₂-n₁n₂ | s=jw

= (s²+8)(s²+3) - 6s.4s | s=jw

= s⁴+11s²+24-24s² | s=jw

= s⁴-13s²+24

= (jw)⁴-13(jw)²+24

⇒ w⁴+13w²+24 ≥ 0

∴ The given function

Z(s) = $\frac{s^2+6s+8}{s^2+4s+3}$ is positive real function

11) $Z(s) = \frac{s^3 + 5s^2 + 9s + 3}{s^3 + 4s^2 + 7s + 9}$

Solution,

Given function is

$$Z(s) = \frac{s^3 + 5s^2 + 9s + 3}{s^3 + 4s^2 + 7s + 9}$$

② Checking for Hurwitz

$$P(s) = s^3 + 5s^2 + 9s + 3$$

Routh's table is

s^3	5	9	0
s^2	3	0	0
s^1	$\frac{15}{2}$	0	
s^0	3		

$P(s)$ is Hurwitz

$$Q(s) = s^3 + 4s^2 + 7s + 9$$

s^3	4	7	0
s^2	9	0	0
s^1	$\frac{19}{4}$	0	
s^0	9		

$Q(s)$ is Hurwitz

③ Hence poles are not lie in imaginary axis so second part is not required.

$$\begin{aligned} m_1 &= 5s^2 + 3 \\ m_2 &= 4s^2 + 9 \\ n_1 &= s^3 + 9s \\ n_2 &= s^3 + 7s \end{aligned}$$

Now,

$$\begin{aligned} m_1 m_2 - n_1 n_2 &|_{s=j\omega} \\ &= (5s^2 + 3)(4s^2 + 9) - (s^3 + 9s)(s^3 + 7s) |_{s=j\omega} \\ &= 20s^4 + 45s^2 + 12s^2 + 27 - s^6 - 7s^4 - 9s^4 - 63s^2 |_{s=j\omega} \\ &= -s^6 + 4s^4 - 6s^2 + 27 |_{s=j\omega} \\ &= -(j\omega)^6 + 4(j\omega)^4 - 6(j\omega)^2 + 27 \\ &\Rightarrow \omega^6 + 4\omega^4 + 6\omega^2 + 27 > 0 \end{aligned}$$

∴ The given function is PRF.

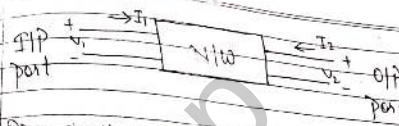
Analysis of two port networks with parameters.

- (i) Open circuit or impedance or Z-parameters.
- (ii) Short circuit or admittance or Y-parameters.
- (iii) Hybrid or h-parameters (combination of Z and Y).
- (iv) Transmission or ABCD parameters.

for analysing two port network

Name of parameters	Dependent variable	Independent variables	Standard Equation
1. Open circuit or impedance or Z-parameters	V_1, V_2	I_1, I_2	$V_1 = Z_{11}I_1 + Z_{12}I_2$ $V_2 = Z_{21}I_1 + Z_{22}I_2$
2. Short circuit or admittance or Y-parameters	I_1, I_2	V_1, V_2	$I_1 = Y_{11}V_1 + Y_{12}V_2$ $I_2 = Y_{21}V_1 + Y_{22}V_2$
3. Hybrid or h-parameters	V_1, I_2	I_1, V_2	$V_1 = h_{11}I_1 + h_{12}V_2$ $I_2 = h_{21}I_1 + h_{22}V_2$
4. ABCD or Transmission	V_1, I_1	V_2, I_2	$V_1 = AV_2 - BI_2$ $I_1 = CV_2 - DI_2$

In analysing two port network, we will assume that such networks are composed of linear elements and contain no independent sources but dependent sources are permissible.



1. Open circuit or impedance or Z-parameters.

characteristics equation $V_1 = Z_{11}I_1 + Z_{12}I_2 \dots \dots \dots (1)$
 $V_2 = Z_{21}I_1 + Z_{22}I_2 \dots \dots \dots (2)$

from equation (1)
When the O/P is opened.
ie $I_2 = 0$

$$V_1 = Z_{11}I_1$$

$$Z_{11} = \frac{V_1}{I_1} \Big|_{I_2=0}$$

from equation (2)
 $V_2 = Z_{21}I_1 + 0$
 $Z_{21} = \frac{V_2}{I_1} \Big|_{I_2=0}$

When I/P is opened
ie $I_1 = 0$

from eqn (1)
 $Z_{12} = \frac{V_1}{I_2} \Big|_{I_1=0}$

from eqn (2)
 $Z_{22} = \frac{V_2}{I_2} \Big|_{I_1=0}$

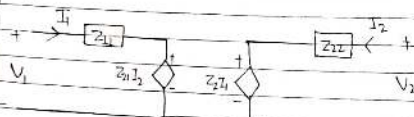
Finally, in matrix form

$$\begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix}$$

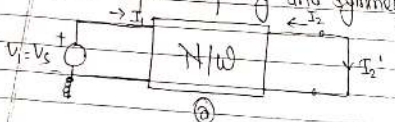
Z_{11} and Z_{22} are known as driving point impedance at input and output respectively.

Z_{12} and Z_{21} are known as reverse and forward transfer impedances respectively.

Equivalent circuit



Condition for reciprocity and symmetry



$$V_1 = Z_{11}I_1 + Z_{12}I_2$$

$$V_2 = Z_{21}I_1 + Z_{22}I_2$$

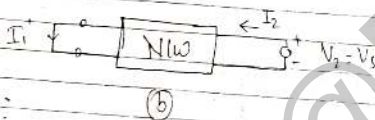


fig a and b are reciprocal if

$$I_1 = I_2'$$

from fig a.

$$V_1 = 0$$

We say $I_2 = -I_2'$ and $V_2 = V_2'$

So, from characteristics equation.

$$V_2 = Z_{11}I_1 - Z_{12}I_2' \quad \text{--- (1)}$$

$$0 = Z_{21}I_1 - Z_{22}I_2' \quad \text{--- (2)}$$

Solving eqn (1) and (2), we get (eliminating I_1)

$$I_2' = \frac{Z_{21}}{Z_{12}Z_{22} - Z_{11}Z_{22}} \times V_2$$

from figure (b)

$$V_2 = V_2', V_1 = 0, I_1 = -I_1'$$

So from characteristics equation

$$0 = -Z_{11}I_1' + Z_{12}I_2 \quad \text{--- (3)}$$

$$V_2 = -Z_{21}I_1' + Z_{22}I_2 \quad \text{--- (4)}$$

Solving (3) and (4) eliminating I_2 , we get

$$I_1' = \frac{Z_{21}}{Z_{11}Z_{22} - Z_{12}Z_{21}} \times V_2$$

Equating $I_1' = I_1$

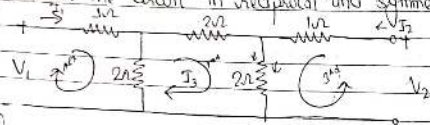
$$\frac{Z_{12}}{Z_{11}Z_{22} - Z_{12}Z_{21}} \times V_2 = \frac{Z_{21}}{Z_{11}Z_{22} - Z_{12}Z_{21}} \times V_1$$

$$\Rightarrow Z_{12} = Z_{21}$$

Symmetry $Z_{11} = Z_{22}$

Analysis of two port networks with Parameters.

Obtain the z-parameters for circuit shown and determine whether the circuit is reciprocal and symmetry.



Solution,

When output port is opened i.e. $I_2 = 0$

Using KVL eqn

$$V_1 = I_1 \times 1 + 2 \times (I_1 - I_2)$$

$$V_1 = 3I_1 - 2I_2 \dots \dots \text{--- (1)}$$

In 2nd mesh

$$2I_2 + 2I_2 + 2(I_2 - I_1) = 0$$

$$\text{Or, } 6I_2 - 2I_1 = 0 \dots \dots \text{--- (2)}$$

And, using 3rd mesh

$$V_2 = 2I_3 \dots \dots \text{--- (3)}$$

Here, $Z_{11} = \frac{V_1}{I_1}$ and $Z_{21} = \frac{V_2}{I_1}$

from eqn (2)

$$6I_3 = 2I_1$$

$$\text{Or, } I_3 = \frac{2I_1}{6}$$

Put I_3 in eqn (1)

$$V_1 = 3I_1 - 2 \times \frac{2I_1}{6}$$

$$V_1 = \frac{14I_1}{6} \Rightarrow \frac{V_1}{I_1} = \frac{7}{3}$$

$$Z_{11} = \frac{V_1}{I_1} = \frac{7}{3} \Omega$$

Put $I_2 = 2I_1$ in eqn (3)

$$V_2 = 2 \times \frac{2I_1}{6}$$

$$V_2 = \frac{4I_1}{6} \Rightarrow \frac{V_2}{I_1} = \frac{2}{3}$$

$$Z_{21} = \frac{V_2}{I_1} = \frac{2}{3} \Omega$$

⇒ When I/P is open ie $I_1=0$

$$V_1 = -2I_3 \dots\dots (4)$$

In 2nd mesh,

$$2I_3 + 2(I_3 + I_2) + 2I_3 = 0$$

$$\text{or, } 6I_3 + 2I_2 = 0 \dots\dots (5)$$

In 3rd mesh,

$$V_2 = I_2 \times 1 + 2(I_2 + I_3)$$

$$V_2 = 3I_2 + 2I_3 \dots\dots (6)$$

Solving equation (4) and (5)

$$V_1 = -2I_3 = -2 \left(-\frac{I_2}{3} \right)$$

$$Z_{12} = \frac{V_1}{I_2} = \frac{2}{3} \Omega$$

from equation (6)

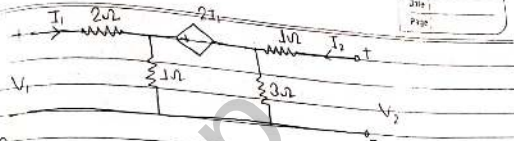
$$V_2 = 3I_2 + \left(-\frac{I_2}{3} \right) \times 2$$

$$V_2 = \frac{7I_2}{3}$$

$$Z_{22} = \frac{V_2}{I_2} = \frac{7}{3} \Omega$$

$$Z_{12} = Z_{21} = \frac{2}{3} \text{ Reciprocal}$$

$$Z_{11} = Z_{22} = \frac{7}{3} \text{ Symmetric}$$



Solution,

Characteristic equations are

$$V_1 = Z_{11}I_1 + Z_{12}I_2 \dots\dots (1)$$

$$V_2 = Z_{21}I_1 + Z_{22}I_2 \dots\dots (2)$$

When O/P current is opened

In first mesh,

$$V_1 = 2I_1 + (I_1 - I_2)$$

$$V_1 = 3I_1 - I_2 \dots\dots (3)$$

2nd mesh,

$$3I_3 + (I_3 - I_1) + 2I_1 = 0$$

$$4I_3 + I_1 = 0 \dots\dots (4)$$

$$V_2 = 3I_3 \dots\dots (5)$$

from eqⁿ (4),

$$I_3 = -\frac{I_1}{4}$$

Eqn ① becomes

$$V_1 = 3I_1 + \frac{I_1}{4} = \frac{13}{4} I_1$$

$$Z_{11} = \frac{V_1}{I_1} = \frac{13}{4} \Omega$$

from equation ③

$$V_2 = 3\left(-\frac{I_1}{4}\right) = -\frac{3}{4} I_1$$

$$Z_{21} = \frac{V_2}{I_1} = -\frac{3}{4} \Omega$$

Now,

When I/P is opened i.e. $I_1 = 0$

$$V_2 = -1 \times I_3 \dots \dots \text{④}$$

2nd mesh,

$$3(I_2 + I_3) + I_3 + 2I_1 = 0$$

$$\text{or } 4I_3 + 3I_2 + 0 = 0$$

$$\Rightarrow 4I_3 + 3I_2 = 0 \dots \dots \text{⑤}$$

In 3rd mesh, $V_2 = I_2 + 3(I_2 + I_3)$

$$V_2 = 4I_2 + 3I_3 \dots \dots \text{⑥}$$

from eqn ⑤

$$4I_3 + 3I_2 = 0$$

$$4I_3 = -3I_2$$

$$\Rightarrow I_3 = -\frac{3}{4} I_2$$

Put I_3 in eqn ⑥

$$V_2 = 4I_2 - \frac{3 \times 3}{4} I_2 = \frac{7}{4} I_2$$

$$\Rightarrow Z_{22} = \frac{V_2}{I_2} = \frac{7}{4} \Omega$$

Also,

Eqn ④

$$V_1 = (-1) \left(-\frac{3}{4}\right) I_2 = \frac{3}{4} I_2$$

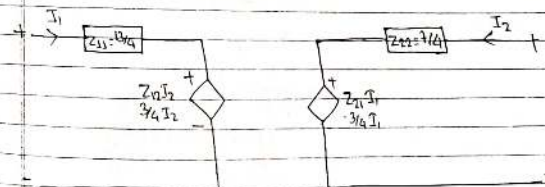
$$\Rightarrow Z_{12} = \frac{3}{4} \Omega$$

In matrix form

$$\begin{bmatrix} \frac{13}{4} & \frac{3}{4} \\ -\frac{3}{4} & \frac{7}{4} \end{bmatrix}$$

$Z_{11} \neq Z_{22}$ not reciprocal
& $Z_{11} \neq Z_{22}$ not symmetric

Equivalent ckt is,



Admittance or Short circuit or Y-parameters

$$I_1 = Y_{11}V_1 + Y_{12}V_2 \dots \text{--- (1)}$$

$$I_2 = Y_{21}V_1 + Y_{22}V_2 \dots \text{--- (2)}$$

$$\begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} Y_{11} & Y_{12} \\ Y_{21} & Y_{22} \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix}$$

↑↑
Y-parameters.

When OP is Shorted ie $V_2 = 0$ from eqⁿ (1)

$$I_1 = Y_{11}V_1 + 0$$

$$Y_{11} = \frac{I_1}{V_1} \Big|_{V_2=0}$$

from eqⁿ (2)

$$I_2 = Y_{21}V_1 + 0$$

$$Y_{21} = \frac{I_2}{V_1}$$

When IP is Shorted ie $V_1 = 0$

from eqⁿ (1)

$$I_1 = 0 + Y_{12}V_2$$

$$Y_{12} = \frac{I_1}{V_2} \Big|_{V_1=0}$$

from eqⁿ (2)

$$I_2 = 0 + Y_{22}V_2$$

$$\Rightarrow Y_{22} = \frac{I_2}{V_2} \Big|_{V_1=0}$$

Y_{11} and Y_{22} are known as driving point admittances at IP and OP respectively.

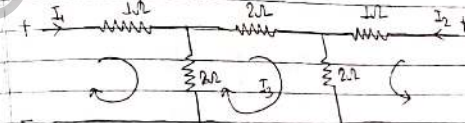
Y_{12} and Y_{21} are reverse and forward transfer admittances respectively.

Condition for Reciprocity and Symmetry

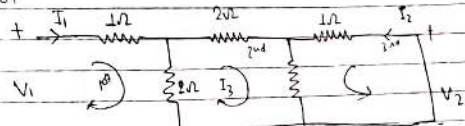
$$Y_{12} = Y_{21} \text{ (Reciprocal)}$$

$$Y_{11} = Y_{22} \text{ (Symmetry)}$$

II Obtain Y-parameters



Solⁿ



Assume OP is Shorted ie $V_1 = 0$

$$V_1 = 1 \times I_1 + 2(I_1 - I_2)$$

$$V_1 = 3I_1 - 2I_2 = 0 \text{ --- (1)}$$

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1. *Journal of the American Medical Association*, 1997; 277: 1033-1038.

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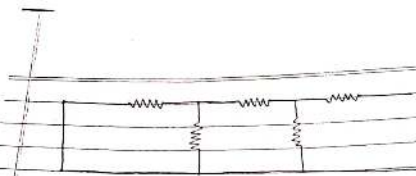
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$$\text{or, } 6I_2 + 2I_2 - 2I_1 = 0$$

$$\text{or, } 3I_2 + I_2 - I_1 = 0 \dots \text{--- (5)}$$

3rd mesh,

$$V_2 = I_2 + 2(I_2 + I_3)$$

$$\text{or, } V_2 = 3I_2 + 2I_3 \dots \text{--- (6)}$$

From eqⁿ (5)

$$3I_2 + I_2 - 2I_3 = 0$$

$$\text{or, } \frac{4}{3}I_2 = I_3 \Rightarrow I_3 = \frac{3}{4}I_2$$

Eqⁿ (6) becomes

$$V_2 = 3I_2 - \frac{3}{4}I_2 = \frac{18}{4}I_2$$

$$\frac{I_2}{V_2} = \frac{V_2}{18} = \frac{1}{18}$$

Also,

from equation (4)

$$I_3 = \frac{3}{2}I_1$$

from eqⁿ (5)

$$3\left(\frac{3}{2}I_1\right) + I_2 - I_1 = 0$$

$$\frac{7}{2}I_1 + I_2 = 0$$

$$\Rightarrow I_2 = -\frac{7}{2}I_1$$

$$V_2 = -\frac{21}{2}I_1 + 3I_1 = -\frac{15}{2}I_1$$

$$\frac{I_1}{V_2} = \frac{V_2}{-15} = -\frac{2}{15}$$

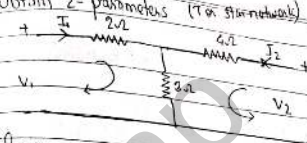
In matrix form

$$\begin{bmatrix} Y_{11} & Y_{12} \\ Y_{21} & Y_{22} \end{bmatrix} = \begin{bmatrix} 1/5\Omega & -2/15 \\ -2/15 & 7/18 \end{bmatrix}$$

$Y_{11} = Y_{21} \rightarrow$ Reciprocal

$Y_{12} \neq Y_{22} \rightarrow$ not symmetric

Obtain Z-parameters (T or star network)



Soln

In 1st mesh,

$$V_1 = 2I_1 + 3(I_1 + I_2)$$

$$V_1 = 5I_1 + 3I_2 \dots \dots \textcircled{1}$$

In 2nd mesh,

$$V_2 = 4I_2 + 3(I_2 + I_1)$$

$$V_2 = 7I_2 + 3I_1 \dots \dots \textcircled{2}$$

Also,

$$V_1 = Z_{11} I_1 + Z_{12} I_2 \dots \dots \textcircled{3}$$

$$V_2 = Z_{21} I_1 + Z_{22} I_2 \dots \dots \textcircled{4}$$

Comparing $\textcircled{1}$ and $\textcircled{3}$

$$Z_{11} = 5\Omega$$

$$Z_{12} = 3\Omega$$

Comparing $\textcircled{2}$ and $\textcircled{4}$

$$Z_{21} = 3\Omega$$

$$Z_{22} = 7\Omega$$

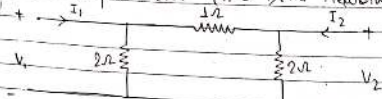
Note: If 2Ω , 4Ω and 3Ω are Z_1 , Z_2 and Z_3 respectively then

$$Z_{11} = Z_1 + Z_3 = 2 + 3 = 5\Omega$$

$$Z_{22} = Z_2 + Z_3 = 4 + 3 = 7\Omega$$

$$Z_{12} = Z_{21} = Z_3 = 3\Omega$$

Obtain Z-parameters (T & Delta network)



Solution,

When ITP is opened $I_3 = 0$ $V_2 = 0$

$$V_1 = 2(I_1 - I_3) \quad \text{--- (i)}$$

$$V_{11} = \frac{3}{2} V$$

$$V_{22} = \frac{3}{2} V$$

$$V_{12} = V_{21} = -1V$$

2nd method,

$$1 \times I_3 + 2(I_2 + I_3) + 2(I_1 - I_3) = 0$$

$$\text{or } I_3 + 2I_2 + 2I_1 - 2I_3 = 0$$

$$\text{or } 5I_3 + 2I_2 - 2I_1 = 0 \quad \text{--- (ii)}$$

3rd method,

$$V_2 = (I_2 + I_3) \cdot 2$$

$$I_2 + I_3 = 0 \quad \text{--- (iii)}$$

Then ITP is shorted $V_1 = 0$

$$V_{11} = \frac{3}{2} V$$

$$V_{22} = \frac{3}{2} V$$

$$V_{12} = V_{21} = -1V$$

$$\Delta \text{ to } Y \rightarrow \frac{R_1 R_2}{R_1 + R_2 + R_3}$$

$$Y \text{ to } \Delta \rightarrow \frac{R_2 + R_3 + R_{31}}{R_3}$$

Another method

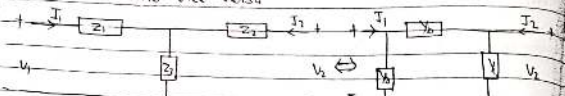
$$Y_a = \frac{1}{2} V, \quad Y_b = \frac{1}{1} \cdot 1V, \quad Y_c = \frac{1}{2} V$$

$$Y_{11} = Y_a + Y_b = \frac{1}{2} + 1 = \frac{3}{2}$$

$$Y_{22} = Y_b + Y_c = \frac{1}{2} + 1 = \frac{3}{2} V$$

$$Y_{12} = Y_{21} = -Y_c = -1$$

Conversion from Star (T) network to Π (Delta) network and vice versa



from T to Π

$$Y_1 = \frac{Z_2 Z_3}{Z_1 + Z_2 + Z_3}$$

$$Y_2 = \frac{Z_1 Z_3}{Z_1 + Z_2 + Z_3}$$

$$Y_3 = \frac{Z_1 Z_2}{Z_1 + Z_2 + Z_3}$$

where $Z_1 = \frac{1}{Y_1}, Z_2 = \frac{1}{Y_2}, Z_3 = \frac{1}{Y_3}$

from Π to T

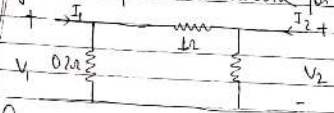
$$Z_1 = \frac{Z_2 Z_3}{Z_1 + Z_2 + Z_3}$$

$$Z_2 = \frac{Z_1 Z_3}{Z_1 + Z_2 + Z_3}$$

$$Z_3 = \frac{Z_1 Z_2}{Z_1 + Z_2 + Z_3}$$

where $Z_1 = \frac{1}{Y_1}, Z_2 = \frac{1}{Y_2}, Z_3 = \frac{1}{Y_3}$

Find the equivalent T network for the given Π network



Solution

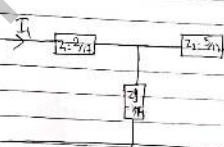
Here, $Y_1 = 0.2 \Omega, Y_2 = 1 \Omega, Y_3 = 0.5 \Omega$

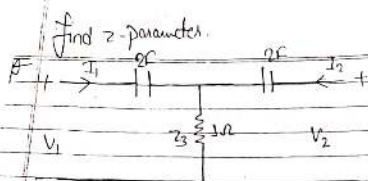
$Z_1 = \frac{1}{0.2} = 5 \Omega, Z_2 = \frac{1}{1} = 1 \Omega, Z_3 = \frac{1}{0.5} = 2 \Omega$

$$Z_1 = \frac{Z_2 Z_3}{Z_1 + Z_2 + Z_3} = \frac{1 \times 2}{1 + 1 + 2} = \frac{2}{4} = 0.5 \Omega$$

$$Z_2 = \frac{Z_1 Z_3}{Z_1 + Z_2 + Z_3} = \frac{0.2 \times 2}{0.2 + 1 + 2} = \frac{0.4}{3.2} = 0.125 \Omega$$

$$Z_3 = \frac{Z_1 Z_2}{Z_1 + Z_2 + Z_3} = \frac{0.2 \times 1}{0.2 + 1 + 2} = \frac{0.2}{3.2} = 0.0625 \Omega$$





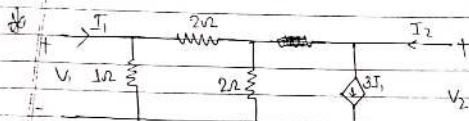
Given
In transformed form,

$$2s = \frac{1}{1/2s}$$

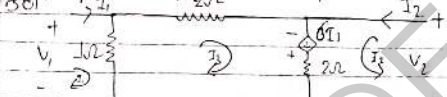
$$Z_{11} = \frac{1}{2s} + 1 = \frac{2s+1}{2s} \Omega$$

$$Z_{22} = \frac{1}{2s} + 1 = \frac{2s+1}{2s} \Omega$$

$$Z_{12} = Z_{21} = 1 \Omega$$



find open ckt parameters (z parameters)



When O/P is opened $I_2 = 0$
 $V_1 = I_1 - I_2 \dots \dots \textcircled{1}$

2nd mesh,

$$2I_3 + 6I_1 + 2I_3 + I_3 - I_1 = 0$$

$$\text{or } 5I_3 + 5I_1 = 0 \quad 5I_3 - 4I_1 = 0 \dots \dots \textcircled{2}$$

$$\text{or } I_3 = I_1 \dots \dots \textcircled{3}$$

3rd mesh,

$$V_1 = 6I_1 + 2I_3 \dots \dots \textcircled{4}$$

Eqn ② becomes

$$V_1 = I_1 + 5I_3 = \frac{14}{5}I_1 - 2\frac{1}{5}I_1$$

$$Z_{11} = \frac{V_1}{I_1} = \frac{12}{5} \Omega - \frac{2}{5} \Omega$$

Eqn ③ becomes

$$V_2 = 6I_1 + 5I_3 + 2 \times \frac{1}{5}I_1$$

$$V_2 = \frac{44}{5}I_1$$

$$\frac{V_2}{I_1} = Z_{21} = \frac{44}{5} \Omega$$

When I/P is opened $I_1 = 0$

$$V_1 = -I_3 \dots \dots \textcircled{4}$$

2nd mesh,

$$2I_3 - 6I_1 + 2(I_3 + I_2) + I_3 = 0$$

$$\begin{aligned} \text{Q1, } 2I_3 - 6I_1 + 2I_3 + 2I_2 + I_3 &= 0 \\ \text{Q1, } 5I_3 - 6I_1 + 2I_2 &= 0 \\ \text{Q1, } 5I_3 + 2I_2 &= 0 \dots \text{--- (5)} \end{aligned}$$

3rd mesh,

$$\begin{aligned} V_3 &= 6I_1 + 2(I_2 + I_3) \\ V_3 &= 2(I_3 + I_2) \dots \text{--- (6)} \end{aligned}$$

Eqⁿ ① becomes

$$V_1 = -\left(\frac{-2}{5} I_2\right)$$

$$\frac{V_1}{I_2} = \frac{2}{5}$$

$$\text{ie } Z_{11} = \frac{V_1}{I_2} = \frac{2}{5} \Omega$$

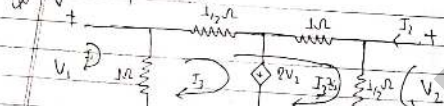
Also,

$$V_2 = 2\left(-\frac{2}{5} I_2 + I_2\right)$$

$$V_2 = \left(\frac{-4}{5} + 2\right) I_2$$

$$\text{ie } Z_{22} = \frac{V_2}{I_2} = \frac{6}{5}$$

Find V_{oc} parameters



Solution,

When OP is shorted $V_2 = 0$

1st mesh,

$$V_1 = I_1 - I_3 \dots \text{--- (1)}$$

2nd mesh,

$$\frac{1}{2} I_3 + 1(I_3 - 2V_1) + \frac{1}{2} I_3 + I_3 = 0$$

$$\frac{1}{2} I_3 + 1(I_3 - 2V_1) + \frac{1}{2} (I_3 - 2V_1 + I_3) + 1(I_3 - I_1) = 0$$

When $V_2 = 0$

$$\frac{I_3}{2} + I_3 + \frac{I_3}{2} + \frac{I_3}{2} + I_3 - I_1 = 0$$

$$\text{Q1, } I_3 + 2I_3 + I_3 + I_3 + 2I_3 - 2I_1 = 0$$

$$\text{Q1, } 6I_3 + I_3 - 2I_1 = 0 \dots \text{--- (2)}$$

3rd mesh,

$$\frac{1}{2} (I_1 + I_3 + 2V_1) = 0$$

$$\text{Q1, } \frac{1}{2} (I_1 + I_3) = 0$$

$$\Rightarrow I_1 + I_3 = 0 \dots \text{--- (3)}$$

Here, eqⁿ ① becomes

$$6I_3 - I_3 - 2I_1 = 0$$

$$\text{Q1, } 5I_3 - 2I_1$$

Eqⁿ ① becomes

$$V_1 = I_1 - \frac{2}{5} I_1 = \frac{3}{5} I_1 \Rightarrow V_{oc} = \frac{5}{3} V$$

Then eqn ④ becomes

$$-6I_1 + I_2 - 2I_3 = 0$$

$$\text{or, } -5I_1 + 2I_3 = 0$$

$$\text{or, } I_1 = -\frac{5}{2} I_3$$

$$V_1 - \frac{5}{2} I_2 + I_2 - \frac{3}{2} I_2$$

$$V_1 - \frac{I_2}{2} = -\frac{2}{3} V$$

When IP is Shorted.

1st mesh,

$$V_1 - I_1(1 - 2I_3) = 0$$

$$\text{or, } I_1 - I_3 = 0 \dots \text{④}$$

In 2nd mesh,

$$\frac{I_3}{2} + \left(\frac{I_3}{2} \cdot 2V_2 \right) + \frac{1}{2} (I_3 - 2V_2 + I_2) + I_3 \times I_1 = 0$$

$$\text{or, } \frac{I_3}{2} + I_3 - 2V_2 + \frac{I_3}{2} - \frac{1}{2} V_2 + \frac{I_2}{2} + I_3 - I_1 = 0$$

$$\text{or, } 3I_3 + \frac{I_2}{2} - 3V_2 - I_1 = 0$$

$$3I_3 + \frac{I_2}{2} - I_1 = 3V_2 \dots \text{⑤}$$

3rd mesh

$$V_2 = \frac{I_2 + I_3}{2} \dots \text{⑥}$$

From eqn ④, $I_1 = I_3$

Then, eqn ⑤ becomes

$$3I_3 + \frac{I_2}{2} - I_3 = 3V_2$$

$$2I_3 + \frac{I_2}{2} = 3V_2$$

$$\text{or, } 2(2V_2 - I_2) + \frac{I_2}{2} = 3V_2$$

$$\text{or, } 4V_2 - 2I_2 + \frac{I_2}{2} = 3V_2$$

$$\text{or, } V_2 - \frac{I_2}{2} = \frac{3}{2} V_2$$

$$\Rightarrow V_2 - \frac{I_2}{2} = -\frac{2}{3} V$$

Then,

$$V_2 = \frac{I_1 + I_2}{2}$$

$$\text{And, } 3I_1 + \frac{I_2}{2} - I_1 = 3V_2$$

$$\text{or, } \frac{2I_1}{2} + \frac{I_2}{2}$$

$$\text{or, } \frac{I_2}{2} = 3V_2 - 2I_1$$

$$\Rightarrow I_2 = 6V_2 - 4I_1$$

$$\text{So, } V_2 = \frac{6V_2 - 4I_1}{2} = 3V_2 - 2I_1$$

$$\therefore 2V_2 = -2I_1$$

$$V_2 - \frac{I_1}{2} = -\frac{2}{3} V$$

Find ABCD parameters or transmission parameters

$$V_2 = AV_1 - BI_1 \quad \text{--- (1)}$$

$$I_2 = CV_1 - DI_1 \quad \text{--- (2)}$$

(Right side is for zero because to V_1, I_1)

$$\begin{bmatrix} V_1 \\ I_1 \end{bmatrix} = \begin{bmatrix} A & B \\ C & D \end{bmatrix} \begin{bmatrix} V_2 \\ -I_2 \end{bmatrix}$$

↑
ABCD parameters

When output is shorted i.e. $V_2 = 0$

From eqn (1)

$$0 = AV_1 - BI_1 \quad | V_2 = 0$$

From eqn (2)

$$0 = CV_1 - DI_1 \quad | V_2 = 0$$

When output is opened

i.e. $I_2 = 0$

From (1) $V_1 = AV_2$

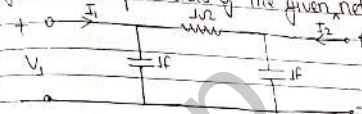
$$A = \frac{V_1}{V_2} \quad | I_2 = 0$$

From (2)

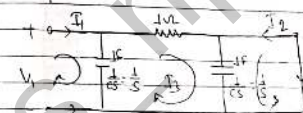
$$I_1 = CV_2$$

$$\Rightarrow C = \frac{I_1}{V_2} \quad | I_2 = 0$$

Find ABCD parameters of the given network



Solution



When output port is shorted circuit i.e. $V_2 = 0$

In 1st mesh,

$$V_1 = 1(I_1 - I_2)$$

$$V_1 = \frac{1}{s} I_1 - \frac{1}{s} I_2 \quad \text{--- (1)}$$

In 2nd mesh,

$$1 \times I_2 + \frac{1}{s}(I_2 + I_1) + \frac{1}{s}(I_2 - I_1) = 0$$

$$\Rightarrow I_2 \left(1 + \frac{1}{s} + \frac{1}{s}\right) + \frac{1}{s} I_1 - \frac{1}{s} I_1 = 0$$

$$\Rightarrow \frac{s+2}{s} I_2 + \frac{1}{s} I_1 - \frac{1}{s} I_1 = 0 \quad \text{--- (2)}$$

In 3rd mesh,

$$\frac{1}{s}(I_2 + I_3) = 0$$

$$0.1 \frac{1}{s} I_2 + \frac{1}{s} I_3 = 0 \dots \textcircled{3}$$

Solving eq^s ①, ② and ③

When O/P port is opened i.e. $I_2 = 0$
In 1st mesh,

$$V_1 = \frac{1}{s} (I_1 - I_3)$$

$$V_1 = \frac{1}{s} I_1 - \frac{1}{s} I_3 \dots \textcircled{4}$$

In 2nd mesh,

$$1 \times I_3 + \frac{1}{s} I_3 + \frac{1}{s} (I_3 - I_1) = 0$$

$$0.1 I_3 \left(1 + \frac{1}{s} + \frac{1}{s} \right) - \frac{1}{s} I_1 = 0$$

$$0.1 I_3 \left(\frac{2+s}{s} \right) - \frac{1}{s} I_1 = 0 \dots \textcircled{5}$$

Again,

$$V_2 = \frac{1}{s} \times I_3 \dots \textcircled{6}$$

$I \rightarrow$ open for $I=0$
 $V \rightarrow$ short for $V=0$

h-parameters

$$V_1 = h_{11} I_1 + h_{12} V_2$$

$$I_2 = h_{21} I_1 + h_{22} V_2$$

$$\begin{bmatrix} V_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{bmatrix} \begin{bmatrix} I_1 \\ V_2 \end{bmatrix}$$

h-parameters

When I/P is opened i.e. $I_2 = 0$ When O/P is shorted i.e. $V_2 = 0$

$$h_{11} = \frac{V_1}{I_1}$$

$$h_{22} = \frac{I_2}{V_2} \quad I_2 = 0$$

$$h_{12} = \frac{V_1}{V_2}$$

$$h_{21} = \frac{I_2}{I_1} \quad V_2 = 0$$

Conversion of parameters

From Y to Z and Z to Y

They are inverse to each other

$$Y \text{ to } Z \quad \begin{bmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{bmatrix} = \begin{bmatrix} Y_{11} & Y_{12} \\ Y_{21} & Y_{22} \end{bmatrix}^{-1} = \frac{1}{\Delta Y} \begin{bmatrix} Y_{22} & -Y_{12} \\ -Y_{21} & Y_{11} \end{bmatrix} \text{ where } \Delta Y = Y_{11}Y_{22} - Y_{12}Y_{21}$$

$$Z \text{ to } Y \quad \begin{bmatrix} Y_{11} & Y_{12} \\ Y_{21} & Y_{22} \end{bmatrix} = \begin{bmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{bmatrix}^{-1} = \frac{1}{\Delta Z} \begin{bmatrix} Z_{22} & -Z_{12} \\ -Z_{21} & Z_{11} \end{bmatrix} \text{ where } \Delta Z = Z_{11}Z_{22} - Z_{12}Z_{21}$$

Conversion between z and h parameters

z-parameters

$$V_1 = Z_{11}I_1 + Z_{12}I_2$$

$$V_2 = Z_{21}I_1 + Z_{22}I_2$$

h-parameters

$$V_1 = h_{11}I_1 + h_{12}V_2 \quad \dots \text{--- (i)}$$

$$I_2 = h_{21}I_1 + h_{22}V_2 \quad \dots \text{--- (ii)}$$

from (i)

$$h_{12}V_2 = I_2 - h_{11}I_1$$

$$V_2 = \frac{I_2}{h_{22}} - \frac{h_{21}}{h_{22}}I_1$$

Put in eqn (ii)

$$V_1 = h_{11}I_1 + h_{12} \left(\frac{I_2}{h_{22}} - \frac{h_{21}}{h_{22}}I_1 \right)$$

$$V_1 = \left(h_{11} - \frac{h_{12}h_{21}}{h_{22}} \right) I_1 + \frac{h_{12}}{h_{22}} I_2$$

Compare with z-parameters

$$Z_{11} = h_{11} - \frac{h_{12}h_{21}}{h_{22}}$$

Conversion between z and h parameters

z-parameters

$$V_1 = Z_{11}I_1 + Z_{12}I_2$$

$$V_2 = Z_{21}I_1 + Z_{22}I_2$$

h-parameters

$$V_1 = h_{11}I_1 + h_{12}V_2 \quad \dots \text{--- (i)}$$

$$I_2 = h_{21}I_1 + h_{22}V_2 \quad \dots \text{--- (ii)}$$

from (i)

$$h_{12}V_2 = I_2 - h_{11}I_1$$

$$V_2 = \frac{I_2}{h_{22}} - \frac{h_{21}}{h_{22}}I_1$$

Put in eqn (ii)

$$V_1 = h_{11}I_1 + h_{12} \left(\frac{I_2}{h_{22}} - \frac{h_{21}}{h_{22}}I_1 \right)$$

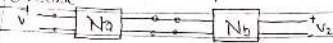
$$V_1 = \left(h_{11} - \frac{h_{12}h_{21}}{h_{22}} \right) I_1 + \frac{h_{12}}{h_{22}} I_2$$

Compare with z-parameters

$$Z_{11} = h_{11} - \frac{h_{12}h_{21}}{h_{22}}$$

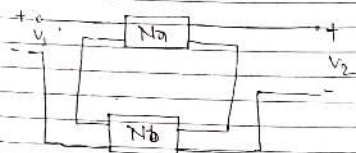
Interconnection of two port networks

① cascade



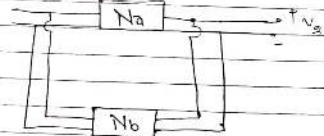
Series-series

② Series-series



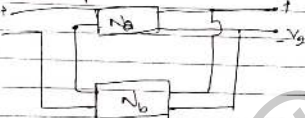
③

parallel-parallel



④

series-parallel



Types of
Interconnection

Individual
Parameter

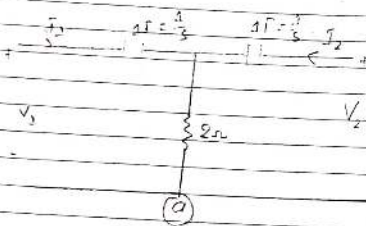
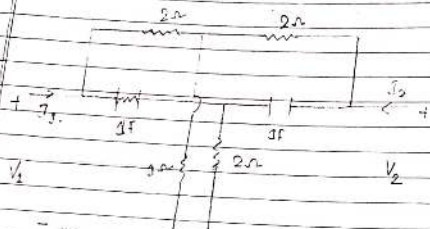
Overall
Parameter

Comments

Types of Interconnection	Individual Parameter	Overall Parameter	Comments
① cascade	$[T_a], [T_b]$	$T = [T_a][T_b]$	ABCD multiply
② Series-series	$[Z_a], [Z_b]$	$Z = [Z_a] + [Z_b]$	Z-addition
③ parallel-parallel	$[Y_a], [Y_b]$	$Y = [Y_a] + [Y_b]$	Y-addition
④ Series-parallel	$[h_a], [h_b]$	$h = [h_a] + [h_b]$	h-addition

Find Y parameters of the ckt

Find T Bridge Network



This is the T-network, so we calculate z-parameters as

$$Z_{11} = \frac{1}{s} + 2 = \frac{2s+1}{s}$$

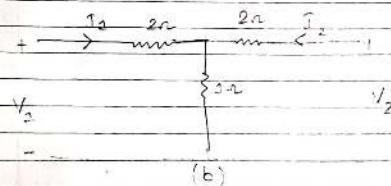
$$Z_{22} = 2 + \frac{1}{s} = \frac{2s+1}{s}$$

$$Z_{12} = Z_{21} = 2\Omega$$

$$\Delta Z = Z_{11}Z_{22} - Z_{12}Z_{21} = \left(\frac{2s+1}{s}\right)^2 - 4 = \frac{4s^2 + 4s + 1 - 4s^2}{s^2} = \frac{4s+1}{s^2}$$

Now, Applying z to Y conversion

$$\begin{bmatrix} Y_{11} & Y_{12} \\ Y_{21} & Y_{22} \end{bmatrix} = \frac{1}{\Delta Z} \begin{bmatrix} Z_{22} & -Z_{12} \\ -Z_{21} & Z_{11} \end{bmatrix} = \frac{1}{\frac{4s+1}{s^2}} \begin{bmatrix} \frac{2s+1}{s} & -2 \\ -2 & \frac{2s+1}{s} \end{bmatrix}$$



This is also T-network, so we calculate Z-parameters.

$$Z_{11} = 2 + 1 = 3, \quad Z_{12} = 1$$

$$Z_{22} = 1 + 2 = 3,$$

$$\Delta Z_b = Z_{11}Z_{22} - Z_{12}Z_{21} =$$

$$= P [8]$$

$$\text{Now, } \begin{bmatrix} Y_{11} & Y_{12} \\ Y_{21} & Y_{22} \end{bmatrix} = \frac{1}{\Delta Z_b} \begin{bmatrix} Z_{22} & -Z_{12} \\ -Z_{21} & Z_{11} \end{bmatrix}$$

$$= \frac{1}{P} \begin{bmatrix} 3 & -1 \\ -1 & 3 \end{bmatrix}$$

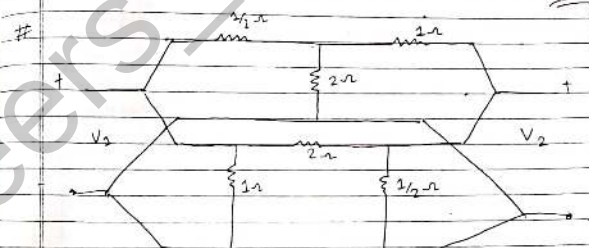
$$= \begin{bmatrix} \frac{3}{P} & -\frac{1}{P} \\ -\frac{1}{P} & \frac{3}{P} \end{bmatrix}$$

$$= \begin{bmatrix} \frac{3}{8} & -\frac{1}{8} \\ -\frac{1}{8} & \frac{3}{8} \end{bmatrix}$$

Total Y-parameters

$$\begin{bmatrix} Y_{11} & Y_{12} \\ Y_{21} & Y_{22} \end{bmatrix} = \begin{bmatrix} Y_{11a} + Y_{11b} & Y_{12a} + Y_{12b} \\ Y_{21a} + Y_{21b} & Y_{22a} + Y_{22b} \end{bmatrix}$$

$$= \begin{bmatrix} \frac{2s+1}{s} + \frac{3}{P} & -\frac{2}{s} - \frac{1}{P} \\ -\frac{2}{s} - \frac{1}{P} & \frac{2s+1}{s} + \frac{3P}{2s} \end{bmatrix}$$



Find Y-parameters of networks?

Soln, $\begin{bmatrix} \frac{2s}{s} & \frac{1}{s} \\ -\frac{2}{s} & -\frac{1}{s} \end{bmatrix}$ (a)

This is T Network, so calculating Z-network

$$Z_{11a} = \frac{1}{2} + 2 = \frac{5}{2} \Omega$$

$$Z_{22a} = 1 + 2 = 3 \Omega$$

$$Z_{21a} = Z_{12a} = 2 \Omega$$

Date _____
Page _____

Page 1

Page 1



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Date _____
Page _____

Page _____



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$$\underbrace{Z_1}_{\text{Z}}$$

$$V_2$$

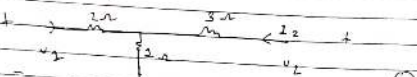
$$v_2$$

4

But $I_1 = -I_2$
 $V_2 = 2I_2 = 2(-I_2)$
 $= -4I_2$

$$Y_{22a} = \frac{I_2}{V_2} = \frac{1}{-4} \Omega$$

$$Y_{21a} = \frac{I_2}{V_1} = \frac{-1}{4} \Omega$$



Network is in T, so finding Z-parameters (b)

$$Z_{11b} = 2 + 1 = 3 \Omega$$

$$Z_{22b} = 3 + 1 = 4 \Omega$$

$$Z_{12b} = Z_{21b} = 1 \Omega$$

$$\Delta Z = Z_{11b}Z_{22b} - Z_{12b}Z_{21b} = 11$$

Converting Z to Y parameters

$$\begin{bmatrix} Y_{11b} & Y_{12b} \\ Y_{21b} & Y_{22b} \end{bmatrix} = \frac{1}{\Delta Z} \begin{bmatrix} Z_{22b} & -Z_{12b} \\ -Z_{21b} & Z_{11b} \end{bmatrix}$$

$$= \frac{1}{11} \begin{bmatrix} 4 & -1 \\ -1 & 3 \end{bmatrix}$$

$$= \begin{bmatrix} \frac{4}{11} & -\frac{1}{11} \\ -\frac{1}{11} & \frac{3}{11} \end{bmatrix}$$

Calculating Y-parameters of network

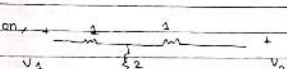
$$\begin{bmatrix} Y_{11} & Y_{12} \\ Y_{21} & Y_{22} \end{bmatrix} = \begin{bmatrix} Y_{11a} & Y_{12a} \\ Y_{21a} & Y_{22a} \end{bmatrix} + \begin{bmatrix} Y_{12a} & Y_{11a} \\ Y_{22a} & Y_{21a} \end{bmatrix}$$

$$= \begin{bmatrix} \frac{23}{44} & -\frac{5}{44} \\ -\frac{5}{44} & \frac{23}{44} \end{bmatrix}$$

Find short ckt parameter. Assume all resistors in Ω



Solution,



(a)

It is in T network, so finding Z parameters

$$Z_{11a} = 1 + 2 = 3 \Omega$$

$$Z_{22a} = 2 + 1 = 3 \Omega$$

$$Z_{12} = 2 \Omega$$

$$\Delta Z = 3 \times 3 - 2 \times 2 = 5$$

Converting Z parameters to Y parameters

$$\begin{bmatrix} Y_{11a} & Y_{12a} \\ Y_{21a} & Y_{22a} \end{bmatrix} = \frac{1}{\Delta Z} \begin{bmatrix} Z_{22a} & -Z_{12a} \\ -Z_{21a} & Z_{11a} \end{bmatrix}$$

$$= \begin{bmatrix} \frac{3}{5} & -\frac{2}{5} \\ -\frac{2}{5} & \frac{3}{5} \end{bmatrix}$$



$$Z_{11b} = 3 \Omega$$

$$Z_{21b} = 3 \Omega$$

$$Z_{12b} = 1 \Omega$$

$$\Delta Z = Z_{11b} Z_{22b} - Z_{21b} Z_{12b}$$

$$= 8 \Omega$$

$$\begin{bmatrix} Y_{11b} & Y_{12b} \\ Y_{21b} & Y_{22b} \end{bmatrix} = \frac{1}{8} \begin{bmatrix} 3 & -1 \\ -1 & 3 \end{bmatrix}$$

$$= \begin{bmatrix} 3/8 & -1/8 \\ -1/8 & 3/8 \end{bmatrix}$$

Finally,

$$\begin{bmatrix} Y_{11} & Y_{12} \\ Y_{21} & Y_{22} \end{bmatrix} = \begin{bmatrix} Y_{11a} + Y_{11b} & Y_{12a} + Y_{12b} \\ Y_{21a} + Y_{21b} & Y_{22a} + Y_{22b} \end{bmatrix}$$

$$= \begin{bmatrix} 39/40 & -21/40 \\ -21/40 & 39/40 \end{bmatrix}$$

The Z-parameters for a two port network are
 $Z_{11} = 2 \Omega$, $Z_{22} = 1 \Omega$, $Z_{12} = Z_{21} = 3 \Omega$
 Compute the transmission parameters of network. Write the network equation using two parameters.

Solution,

$$V_1 = Z_{11} I_1 + Z_{12} I_2 \quad \dots (1)$$

$$V_2 = Z_{21} I_1 + Z_{22} I_2 \quad \dots (2)$$

$$V_1 = A V_2 + B I_2 \quad \dots (3)$$

$$I_1 = C V_2 + D I_2 \quad \dots (4)$$

From (2)

$$Z_{21} I_1 = V_2 - Z_{22} I_2$$

$$I_1 = \frac{V_2 - Z_{22} I_2}{Z_{21}} = \frac{V_2}{Z_{21}} - \frac{Z_{22}}{Z_{21}} I_2 \quad \dots (5)$$

Put in (1)

$$V_1 = Z_{11} \left(\frac{V_2}{Z_{21}} - \frac{Z_{22}}{Z_{21}} I_2 \right) + Z_{12} I_2$$

$$\text{or } V_1 = \frac{Z_{11} V_2}{Z_{21}} + \left(-\frac{Z_{11} Z_{22}}{Z_{21}} + Z_{12} \right) I_2 \quad \dots (6)$$

Comparing (3) & (6)

$$A = \frac{Z_{11}}{Z_{21}} = \frac{2}{3}, \quad B = \frac{Z_{11} Z_{22}}{Z_{21}} + Z_{12} = \frac{20}{3}$$

Comparing (4) & (5)

$$C = \frac{1}{Z_{21}} = \frac{1}{3}, \quad D = \frac{Z_{22}}{Z_{21}} = \frac{10}{30} = \frac{1}{3}$$

$$\text{Finally } \frac{2}{3} V_2 + \frac{20}{3} I_2$$

$$V_1 = \frac{2}{3} V_2 + \frac{20}{3} I_2 \quad \text{or } \text{reqd. eqn.}$$

$$I_1 = \frac{1}{3} V_2 - \frac{1}{3} I_2$$

Date _____
Page _____

Page |

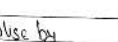
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Page _____



star B form

network.

- network.



use by

Ad 1

$$\langle S^2 + 1 \rangle$$

$$2s + 1$$

B + D)



Inductor - coefficient of s in numerator
constant in denominator

Capacitor - coefficient of $\frac{1}{s}$ in numerator

$$Z(s) = \frac{2(s^2+1)(s^2+5)}{s(s^2+4)}$$

Realize in Foster 1st form.

Soln

Partial fraction.

$$\frac{s^5+5s^3+2s^2+18}{2s^4+8s^2}$$

$$= \frac{12s^2+18}{2s^4+8s^2}$$

$$Z(s) = 2s + \frac{12s^2+18}{s(s^2+4)}$$

Taking partial fraction

$$\frac{12s^2+18}{s(s^2+4)} = \frac{A}{s} + \frac{Bs+C}{s^2+4}$$

$$= \frac{A(s^2+4) + s(Bs+C)}{s(s^2+4)}$$

$$= \frac{As^2+4A+Bs^2+Cs}{s(s^2+4)}$$

$$= \frac{(A+B)s^2 + Cs + 4A}{s(s^2+4)}$$

Equating the coefficient

$$A+B=12$$

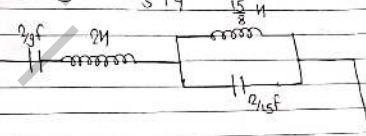
$$C=0$$

$$4A=18$$

$$\Rightarrow A=3\frac{1}{2}$$

$$B=\frac{15}{2}$$

$$Z(s) = 2s + \frac{9/2}{s} + \frac{15/2 s}{s^2+4}$$



fastest 2nd form (parallel form)



* $Z(s) = \frac{s(s^2+2)}{(s^2+1)(s^2+3)}$ Realize in fastest 1st and 2nd form.

Solution

$$Y(s) = \frac{(s^2+1)(s^2+3)}{s(s^2+2)}$$

$$\frac{s^4+2s^2+3}{s^4+2s^2}$$

$$2s^2+3$$

$$= s + \frac{2s^2+3}{s(s^2+2)}$$

Taking $\frac{2s^2+3}{s(s^2+2)} = \frac{A}{s} + \frac{Bs+C}{s^2+2}$

$$= \frac{A(s^2+2) + s(Bs+C)}{s(s^2+2)}$$

$$\frac{2s^2+3}{s(s^2+2)} = \frac{(A+B)s^2 + Cs + 2A}{s(s^2+2)}$$

Equating coefficients

$$A+B=2$$

$$C=0$$

$$2A=3$$

$$\Rightarrow A = \frac{3}{2}$$

$$B = \frac{1}{2}$$

$$Y(s) = s + \frac{3/2}{s} + \frac{1/2 s + 0}{s^2+2}$$

LC Network is



* Cover 1st form (ladder form)



Continue Division method

* $Z(s) = \frac{s(s^2+2)}{(s^2+1)(s^2+3)}$

* $Z(s) = \frac{2(s^2+1)(s^2+3)}{s(s^2+4)}$

Solution

$$Z(s) = \frac{2(s^2+1)(s^2+3)}{s(s^2+4)}$$

$$= \frac{2(s^4+4s^2+3)}{s^4+4s^2}$$

$$= \frac{2s^4+8s^2+6}{s^4+4s^2}$$

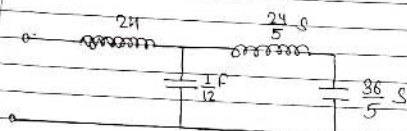
Put in descending order

$$\begin{array}{r} s^3 + 4s \\ 2s^4 + 20s^2 + 18 \end{array} \quad \begin{array}{r} 2s \\ 2s^4 + 8s^2 \end{array}$$

$$\begin{array}{r} 12s^2 + 18 \\ s^3 + 4s \end{array} \quad \begin{array}{r} 3/12 \\ s^3 + 18s \end{array}$$

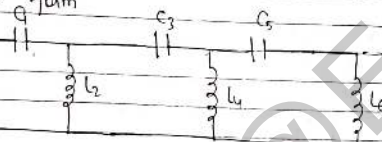
$$\begin{array}{r} 5s \\ 12s^2 + 18 \end{array} \quad \begin{array}{r} 24/5s \\ 12s^2 + 18 \end{array}$$

$$\begin{array}{r} 18 \\ 5s \end{array} \quad \begin{array}{r} 36/5s \\ 5s \end{array}$$



Note: first inductor second capacitor values are same for inductor & capacitor.

Conver and form



Note: all values are reversed

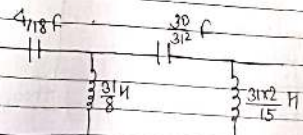
$$\begin{array}{r} 2(s^2+1)(s^2+9) \\ s(s^2+4) \\ 2s^4 + 20s^2 + 18 \\ s^3 + 4s \\ 18 + 20s^2 + 2s^4 \\ 4s + s^3 \end{array} \Rightarrow \text{Arrange in ascending order}$$

$$\begin{array}{r} 4s + s^3 \\ 18 + 20s^2 + 2s^4 \end{array} \quad \begin{array}{r} 38 \\ 45 \end{array} \quad \begin{array}{r} 18 + 20s^2 + 2s^4 \\ 18 + 9/2s^2 \end{array}$$

$$\begin{array}{r} 31s^2 + 2s^4 \\ 4s + s^3 \end{array} \quad \begin{array}{r} 8 \\ 31s \end{array} \quad \begin{array}{r} 4s + s^3 \\ 4s + 16/31s^3 \end{array}$$

$$\begin{array}{r} 15s^3 \\ 31 \end{array} \quad \begin{array}{r} 31s^2 + 2s^4 \\ 31 \end{array} \quad \begin{array}{r} 81 \\ 30s \end{array}$$

$$\begin{array}{r} 2s^4 \\ 15s^3 \\ 31 \end{array} \quad \begin{array}{r} 15 \\ 31x2s \end{array}$$

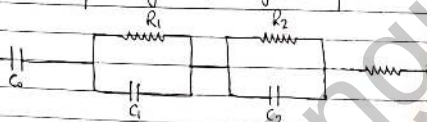


RC impedance functions or RL admittance function

Properties of RC driving point impedance function

- All the poles and zeros are simple and are located on the negative real axis
- Poles and zeros are interlaced
- The singularity nearest to (or at) the origin is a pole.
- The singularity nearest to (or at) the infinity is a zero.
- The residues of a pole of $Z(s)$ are real and positive.

At infinity nearest infinity zero
At origin or near origin is a pole



If $Z(s) = \frac{k_0}{s} + \frac{k_1}{s+\sigma_1} + \frac{k_2}{s+\sigma_2} + \dots + \frac{k_i}{s+\sigma_i} + \dots + k_{\infty}$ then,

$C_0 = \frac{1}{k_0}$, $R_{\infty} = k_{\infty}$, $C_1 = \frac{1}{-k_1}$, $R_1 = \frac{k_1}{\sigma_1}$, $R_2 = \frac{k_2}{\sigma_2}$, $C_2 = \frac{1}{k_2}$

Ex $F(s) = \frac{2(s+1)(s+3)}{s(s+2)}$ Realize in Foster 1st form

Solution

Here All zeros and poles lie in negative real axis so it is RC impedance function (RL admittance function)

$$F(s) = \frac{2s^2 + 4s + 3}{s^2 + 2s}$$

$$= \frac{2s^2 + 8s + 6}{s^2 + 2s} - \frac{4s + 3}{s^2 + 2s}$$

$$F(s) = 2 + \frac{4s+6}{s(s+2)}$$

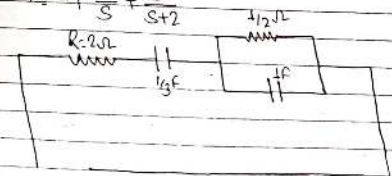
Taking $\frac{4s+6}{s(s+2)} = \frac{A}{s} + \frac{B}{s+2}$

$$4s+6 = A(s+2) + Bs$$

On $4s+6 = (A+B)s + 2A$

$\Rightarrow A=3$ & $B=1$

$$F(s) = 2 + \frac{3}{s} + \frac{1}{s+2}$$



Ex. $F(s) = \frac{2(s+1)(s+3)}{s(s+2)}$ Realize in Foster 1st form

If $F(s)$ represents an admittance function, then it may be realized as an RC admittance function

$$F(s) = Y_{RC}(s) = \frac{2(s+1)(s+3)}{s(s+2)}$$

For Foster 2nd form



Resistor is in series and value of capacitor in 1st form is value of inductor in 2nd form

Chapter 9: Two Port Passive Network



Four variables are available for the study of the network (V_1, V_2, I_1, I_2)

$V_1, I_1 \rightarrow$ independent variable

$V_2, I_2 \rightarrow$ dependent variable (depends on the problem under consideration, or parameters required).



T-network

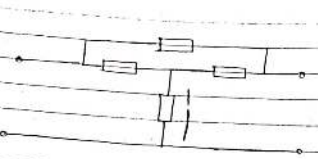
pi-network



Ladder network



L-network



Bridge T network

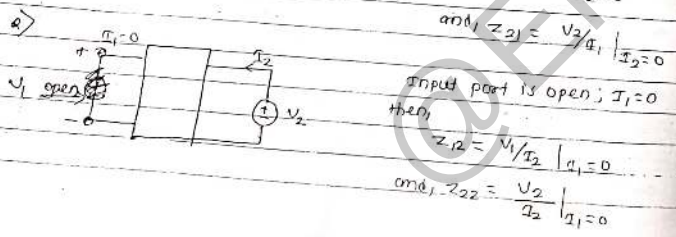
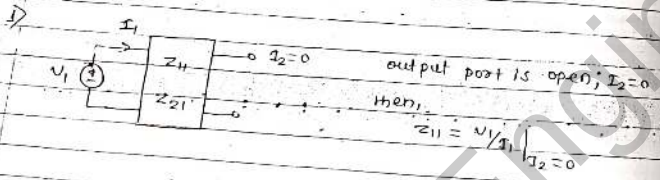
* two port network parameters:-

(i) open ext impedance (z) parameters:-

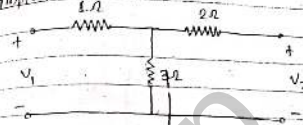
In z-parameter representation of network, network is analyzed in terms of four impedances,

$z_{11}, z_{12}, z_{21}, z_{22}$	z_{11} - driving point impedance
Defining equations are;	z_{12} - transfer impedance
$V_1 = z_{11}I_1 + z_{12}I_2$	z_{21} - reverse transfer impedance
$V_2 = z_{21}I_1 + z_{22}I_2$	z_{22} - output impedance

calculation:-



Example:-



Find z-parameters:

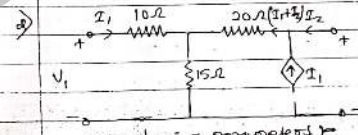
soln,

$$z_{11} = \frac{V_1}{I_1} \Big|_{I_2=0} \quad z_{21} = \frac{V_2}{I_1} \Big|_{I_2=0}$$

$$= \frac{4I_1}{I_1} = 4\Omega \quad = \frac{3I_1}{I_1} = 3\Omega$$

$$z_{12} = \frac{V_1}{I_2} \Big|_{I_1=0} \quad z_{22} = \frac{V_2}{I_2} \Big|_{I_1=0}$$

$$= \frac{3I_2}{I_2} = 3\Omega \quad = \frac{(2+3)I_2}{I_2} = 5\Omega$$



Determine z parameters

soln,

Using KVL,
 $V_1 = 10I_1 + 15(2I_1 + I_2)$
 $\therefore V_1 = 40I_1 + 15I_2$

$$V_2 = 20(I_1 + I_2) + 15(2I_1 + I_2)$$

$$\therefore V_2 = 50I_1 + 35I_2$$

Now, z-parameters;

$$Z_{11} = \frac{V_1}{I_1} \bigg|_{I_2=0}$$

$$= \frac{40I_1 + 15(0)}{I_1} = 40\Omega$$

$$Z_{21} = \frac{V_2}{I_1} \bigg|_{I_2=0} = \frac{50I_1 + 25(0)}{I_1} = 50\Omega$$

$$Z_{12} = \frac{V_1}{I_2} \bigg|_{I_1=0} = \frac{40(0) + 15I_2}{I_2} = 15\Omega$$

$$\therefore Z_{22} = \frac{V_2}{I_2} \bigg|_{I_1=0} = \frac{50(0) + 25I_2}{I_2} = 25\Omega$$

(III) short circuit admittance parameter (Y-parameter):

In admittance (Y) representation, V_1, I_1, V_2 & I_2 in two port network are related through admittances.

$$(I_1, I_2) = f(V_1, V_2)$$

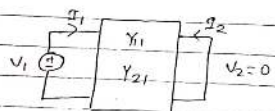
The defining relationships are;

$$I_1 = Y_{11}V_1 + Y_{12}V_2$$

$$I_2 = Y_{21}V_1 + Y_{22}V_2$$

* Calculation:

(i)

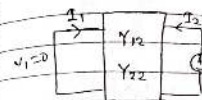


output port is short circuited, ($V_2=0$)

$$Y_{11} = \frac{I_1}{V_1} \bigg|_{V_2=0}$$

$$Y_{21} = \frac{I_2}{V_1} \bigg|_{V_2=0}$$

(ii)



Input port is shorted ($V_1=0$)

$$Y_{12} = \frac{I_1}{V_2} \bigg|_{V_1=0}$$

$$Y_{22} = \frac{I_2}{V_2} \bigg|_{V_1=0}$$

$$Y_{11} = \frac{I_1}{V_1} \bigg|_{V_2=0}$$

Y_{11} - o/p driving point admittance

Y_{12} - transfer admittance

Y_{21} - reverse transfer admittance

Y_{22} - o/p driving point admittance.

Example



So in,

$$V_1 = 4I_1 + 3I_2 \quad \text{--- (1)}$$

$$V_2 = 3I_1 + 5I_2 \quad \text{--- (2)}$$

$$Y_{11} = \frac{I_1}{V_1} \bigg|_{V_2=0}$$

$$= \frac{I_1}{\frac{4}{3}I_1} = \frac{3}{4}S$$

$$\text{From (1), } V_1 = 4I_1 - \left(\frac{3}{5}\right)I_1 \times 3$$

$$= \frac{11}{5}I_1$$

$$I_1 = -\frac{9}{11}I_2$$

$$Y_{21} = \frac{I_2}{V_1} \bigg|_{V_2=0}$$

$$= \frac{I_2}{\frac{11}{5}I_1} = \frac{5}{11} \times \left(-\frac{9}{11}\right) = -\frac{45}{121}S$$

$$= -\frac{45}{121}S$$

$$= -\frac{45}{121}S$$

$$Y_{12} = \left. \frac{I_1}{V_2} \right|_{V_1=0}$$

$$= \frac{I_1}{V_2}$$

$$= -\frac{1}{11} \frac{I_1}{V_2}$$

$$= -\frac{1}{11} \frac{I_1}{V_2}$$

$$Y_{22} = \left. \frac{I_2}{V_2} \right|_{V_1=0}$$

$$= \frac{I_2}{V_2}$$

$$= \frac{1}{4} \frac{I_2}{V_2}$$

$$= \frac{1}{4} \frac{I_2}{V_2}$$

(III) Hybrid parameters:-

Hybrid parameters relate the voltage at the input port and current at the o/p port with the current at the input port & voltage at output port,

$$(V_1, I_2) = f(I_1, V_2)$$

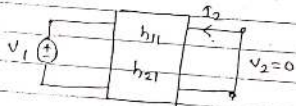
Defining relationships;

$$V_1 = h_{11} I_1 + h_{12} V_2$$

$$I_2 = h_{21} I_1 + h_{22} V_2$$

* calculations;

(I)

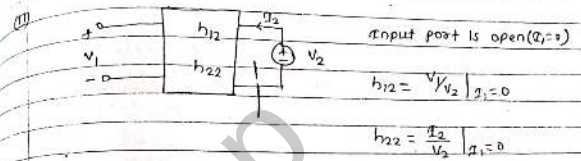


output is short circuited;

$$V_2 = 0$$

$$h_{11} = \left. \frac{V_1}{I_1} \right|_{V_2=0}$$

$$h_{21} = \left. \frac{I_2}{I_1} \right|_{V_2=0}$$



input port is open ($I_1=0$)

$$h_{12} = \left. \frac{V_1}{V_2} \right|_{I_1=0}$$

$$h_{22} = \left. \frac{I_2}{V_2} \right|_{I_1=0}$$

(IV) Transmission (A, B, C, D) parameters (T-parameters):-

In this case, input port variable (V_1, I_1) are expressed in terms of output port variables ($V_2, -I_2$)

Defining relations:-

$$V_1 = A V_2 + B (-I_2)$$

$$I_1 = C V_2 + D (-I_2)$$

calculations:-

(I)

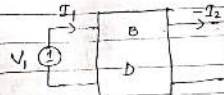


output port is open ($I_2=0$)

$$A = \left. \frac{V_1}{V_2} \right|_{I_2=0}$$

$$C = \left. \frac{I_1}{V_2} \right|_{I_2=0}$$

(II)



output port is shorted ($V_2=0$)

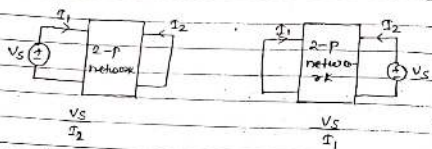
$$B = \left. \frac{V_1}{-I_2} \right|_{V_2=0}$$

$$D = \left. \frac{I_1}{-I_2} \right|_{V_2=0}$$

Reciprocity & symmetry: SHORT NOTE

Reciprocity:-

A two-port network is said to be reciprocal if the ratio of excitation to response is not changed with the interchange of their position.



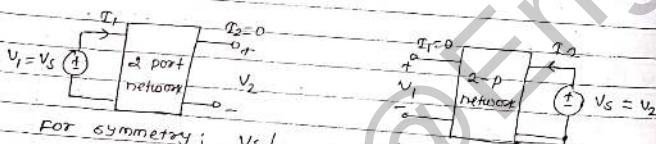
For reciprocity, $\frac{V_2}{I_1} = \frac{V_1}{I_2}$

For any reciprocal network;

$$\left. \begin{aligned} Z_{12} &= Z_{21} \\ Y_{12} &= Y_{21} \\ AD - BC &= 1 \\ h_{12} &= -h_{21} \end{aligned} \right\} \text{ must be present/determined}$$

Symmetry:-

A two port network is said to be symmetrical, if the ports can be interchanged without changing the port voltages and currents.



For symmetry; $\frac{V_2}{I_1} \bigg|_{I_2=0} = \frac{V_1}{I_2} \bigg|_{I_1=0}$

In any symmetrical network;

$$Z_{11} = Z_{22}$$

$$Y_{11} = Y_{22}$$

$$A = D$$

$$h_{11}h_{22} - h_{12}h_{21} = 1$$

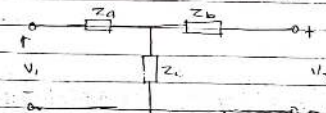
$$\begin{vmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{vmatrix} = 1$$

'T' & 'π' Networks:-

Symmetrical networks are generally represented in the form of 'T' or 'π' networks. Equivalent ckt. of transmission lines and complex filter circuits. also either in 'T' or 'π' form.

T-network:-

The T-network is represented by the model shown in fig. Z-parameters are used to determine components.



T-network components in the form of, Z parameters;

$$Z_{11} = Z_a + Z_c \quad \text{--- (1)}$$

$$Z_{12} = Z_{21} = Z_c \quad \text{--- (2)}$$

$$Z_{22} = Z_b + Z_c \quad \text{--- (3)}$$

For any network, Z_a, Z_b & Z_c are given by,

$$Z_a = Z_{11} - Z_{12}$$

$$\begin{cases} Z_{11} = Z_{22} - Z_{12} \\ Z_c = Z_{12} = Z_{21} \end{cases}$$

so if Z-parameters of a network are given, it's convenient to represent it in π -form.

π -network:-

A π -network is arranged as shown in the fig.

Y-parameters of

a π -network;

$$Y_{11} = Y_a + Y_c$$

$$Y_{12} = Y_c$$

$$Y_{22} = Y_b + Y_c$$

so,

$$Y_a = Y_{11} - Y_{12}$$

$$Y_b = Y_{22} - Y_{12}$$

$$Y_c = -Y_{12}$$

If Y-parameters of a network are known; it's convenient to represent it in π -network.

Find the output equivalent π -network for the given π -network.

soln,

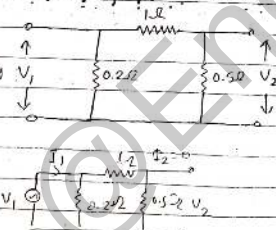
for π -network, Z-parameters

are needed

$$Z_{11} = \left. \frac{V_1}{I_1} \right|_{I_2=0}$$

$$= \frac{1.2 \times 0.2}{0.2 + 1.5} = 0.17 \Omega$$

$$= 0.17 \Omega$$



$$Z_{21} = \left. \frac{V_2}{I_1} \right|_{I_2=0} = \frac{1.2 \times 0.2}{1.7} = \frac{1}{17} \Omega$$

$$Z_{12} = \left. \frac{V_1}{I_2} \right|_{I_1=0}$$

$$\text{where } V_1 = I_2 \times 0.5$$

$$\Rightarrow Z_{12} = \frac{I_2 \times 0.5}{I_2} = 0.5 \times \frac{0.2}{1.7} = \frac{1}{17} \Omega$$

$$Z_{22} = \left. \frac{V_2}{I_2} \right|_{I_1=0}$$

$$\text{where } V_2 = I_2 \times \frac{1.2 \times 0.5}{1.7}$$

$$\Rightarrow Z_{22} = \frac{1.2 \times 0.5}{1.7} = \frac{6}{17} \Omega$$

Now,

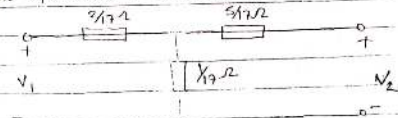
$$Z_a = Z_{11} - Z_{12}$$

$$= \frac{3}{17} - \frac{1}{17} = \frac{2}{17} \Omega$$

$$Z_b = Z_{22} - Z_{12} = \frac{6}{17} - \frac{1}{17} = \frac{5}{17} \Omega$$

$$Z_c = Z_{12} = \frac{1}{17} \Omega$$

The equivalent π -network is,

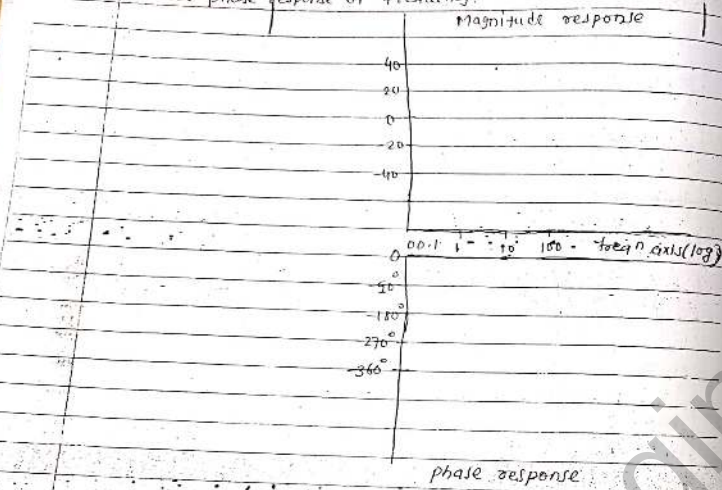


$$\text{decibel value of } b = 20 \log b \\ = 15.56$$

Bode diagrams:-

A bode plot is used to show the frequency response of a network function.

It has two graphs; one shows the magnitude response & other phase response of frequency.



Bode diagram gives the basic trend of a network function with change in freq.

Let a typical transfer function be given by

$$G(j\omega) = K(1+j\omega T_a)(1+j\omega T_b) \\ (j\omega)^r(1+j\omega T_1)(1+j\omega T_2)(1+2\zeta\omega_n+\omega_n^2)$$

decibel value of this function is,

$$20 \log |G(j\omega)| = 20 \log K + 20 \log(1+j\omega T_a) + \dots$$

* process of constructing a bode diagram:-

- 1) identify corner freq. (frequency for zero & pole),
- 2) start drawing from the magnitude & phase for low frequency ($\omega = 0.1$)

3) For magnitude plot;

- 20 dB/decade for a simple pole.

(-40 dB/decade for pole of multiplicity 2).

+ 20 dB/decade for a simple zero.

+ 40 dB/decade for a zero of multiplicity 2 & 40 dB/decade for a complex zero/pole.

+/- 40 dB/decade for a complex zero/pole of multiplicity 2.

For phase:-

-90°/decade for a pole at zero origin.

+90°/decade for a zero at zero origin.

-45°/decade for a simple pole.

+45°/decade for a simple zero.

+/- 90°/decade for complex conjugate poles & zeros.

$$G_2(s) = \frac{10(s+10)}{s(s+2)(s+5)}$$

$$G_2(s) = \frac{10(s+10)}{s(s+2)(s+5)}$$

515

This function has three poles at $s=0$, $s=-a$, $s=-s$ and, zero at $s=-10$

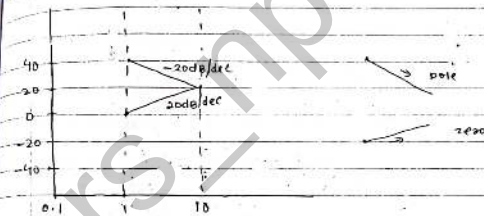
For the starting point of magnitude plot;
when $\omega = 0.1 \text{ rad/s}$.

$$20 \log |G(j\omega)| \approx 20 \log \frac{10}{0.1} = 40 \text{ dB}$$

$$G(j\omega) = \frac{10(j\omega + j0)}{j\omega(j\omega + 2)(j\omega + 5)}$$

$$= \frac{10(1 + j0 \cdot 1\omega)}{j\omega(1 + j0 \cdot 2\omega)(1 + j0 \cdot 5\omega)}$$

$$= \frac{10(1 + j0.1\omega)}{j\omega(1 + j0.5\omega)(1 + j0.2\omega)}$$



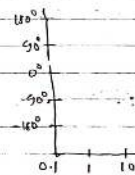
4) For second order equation, write it in the form of $s^2 + 2\zeta\omega_n s + \omega_n^2$ and determine ω_n .

draw Bode diagram with $\pm 40 \text{ dB/dec}$ at the corner frequency,

o) phase plot:-

4) with ; no at origin, start from -90°

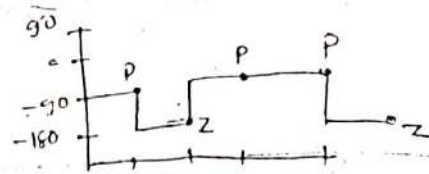
11 zero 11 11 11 11 +90°



If no pole or zero at origin,
start from phase at $\omega \rightarrow 0$.

(w at least one degree lower than the lowest corner temp).

2) Straight line until next corner frequency. Then jump down -90° for a pole (or up for $+90^\circ$ for a zero).



3) Repeat 2) for all simple poles & zeros.

4) $\pm 180^\circ$ for a complex zero or pole

5) connect mid-points of all phase jumps

$G(s) = \frac{10(s+10)}{s(s+2)(s+5)}$
~~so, it is a complex zero~~
 replace s by $j\omega$

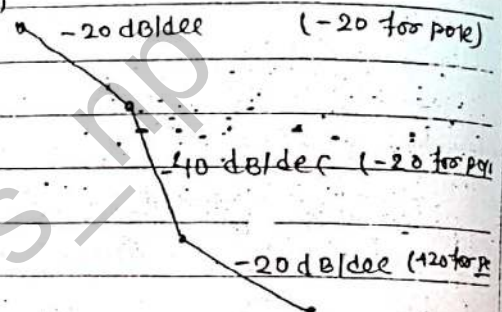
⇒ Rough

$s = 0$ & $s = -2 \Rightarrow |s| = 2$

∴ poles are 0 & 2

and zero is 10

then,



so, it,

corner frequencies are 2, 5 & 10

replace s by $j\omega$,

let $\omega = 0.1$

then,

$G(j\omega) = \frac{10(0.1j + 10)}{0.1j(0.1j + 2)(0.1j + 5)} \approx \frac{10}{0.1j}$

then,

starting point = $20 \log \left(\frac{10}{0.1} \right) = 40$